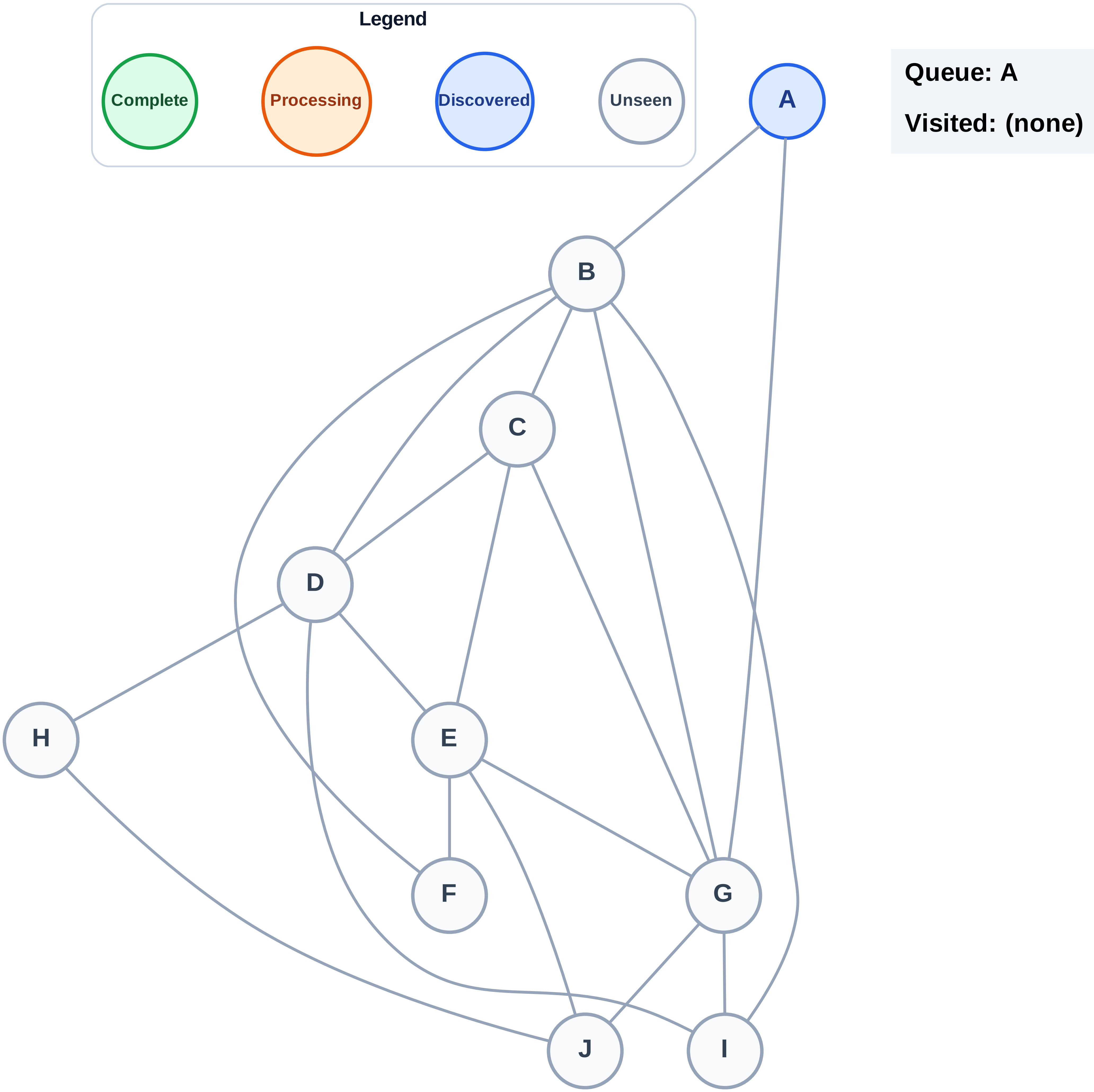
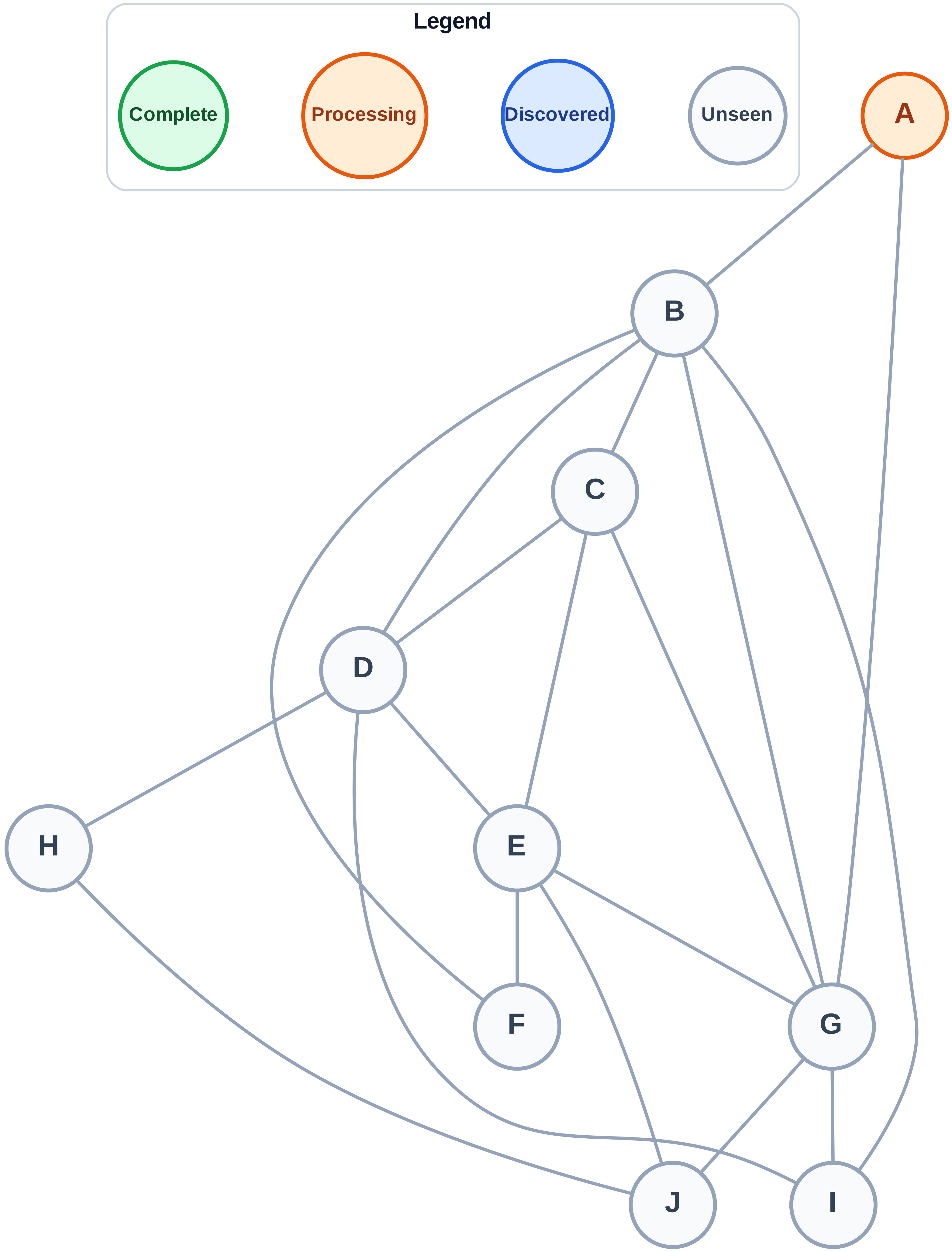
BFS Traversal

Step 1: Start at A



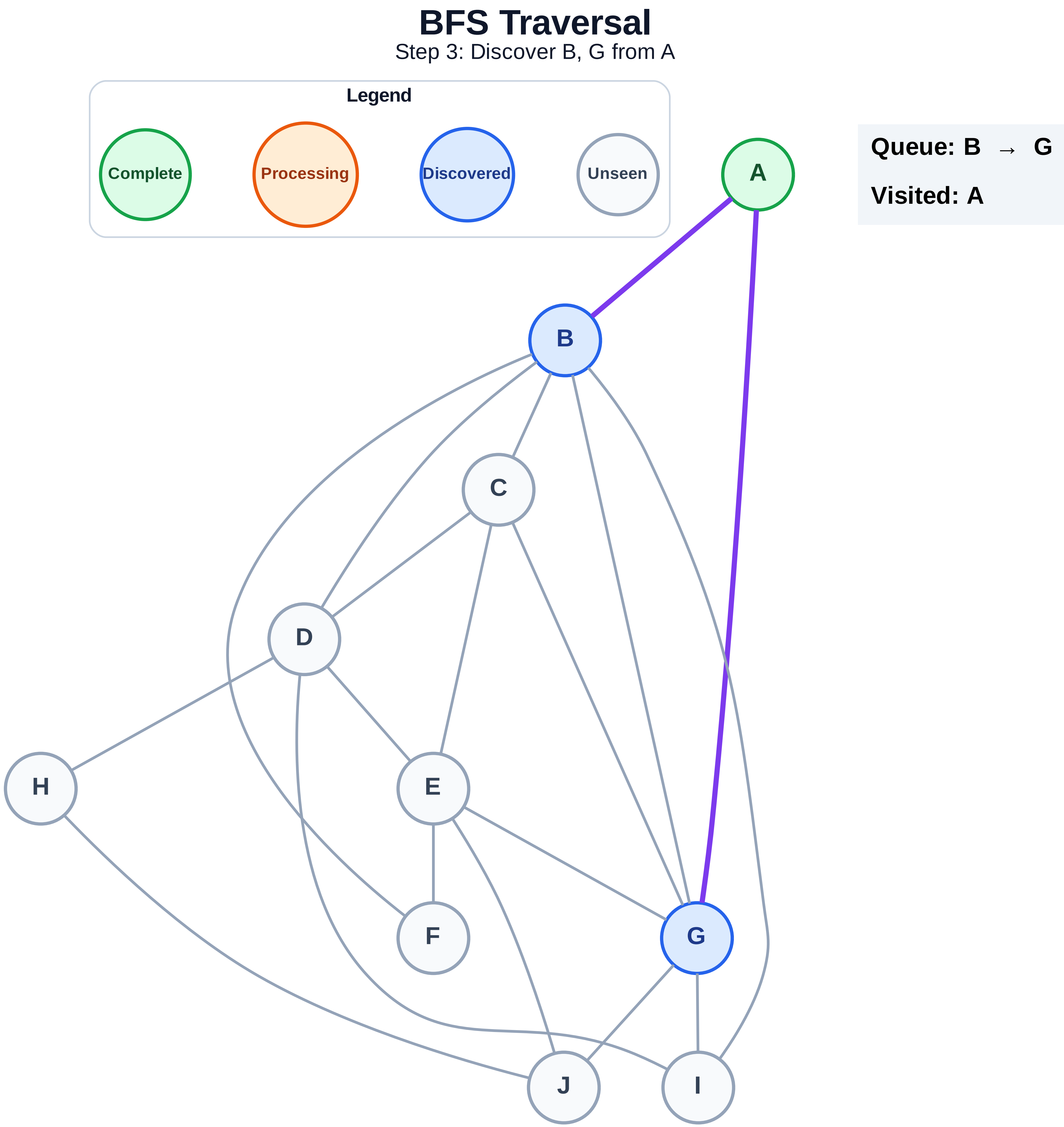
BFS Traversal

Step 2: Process node A



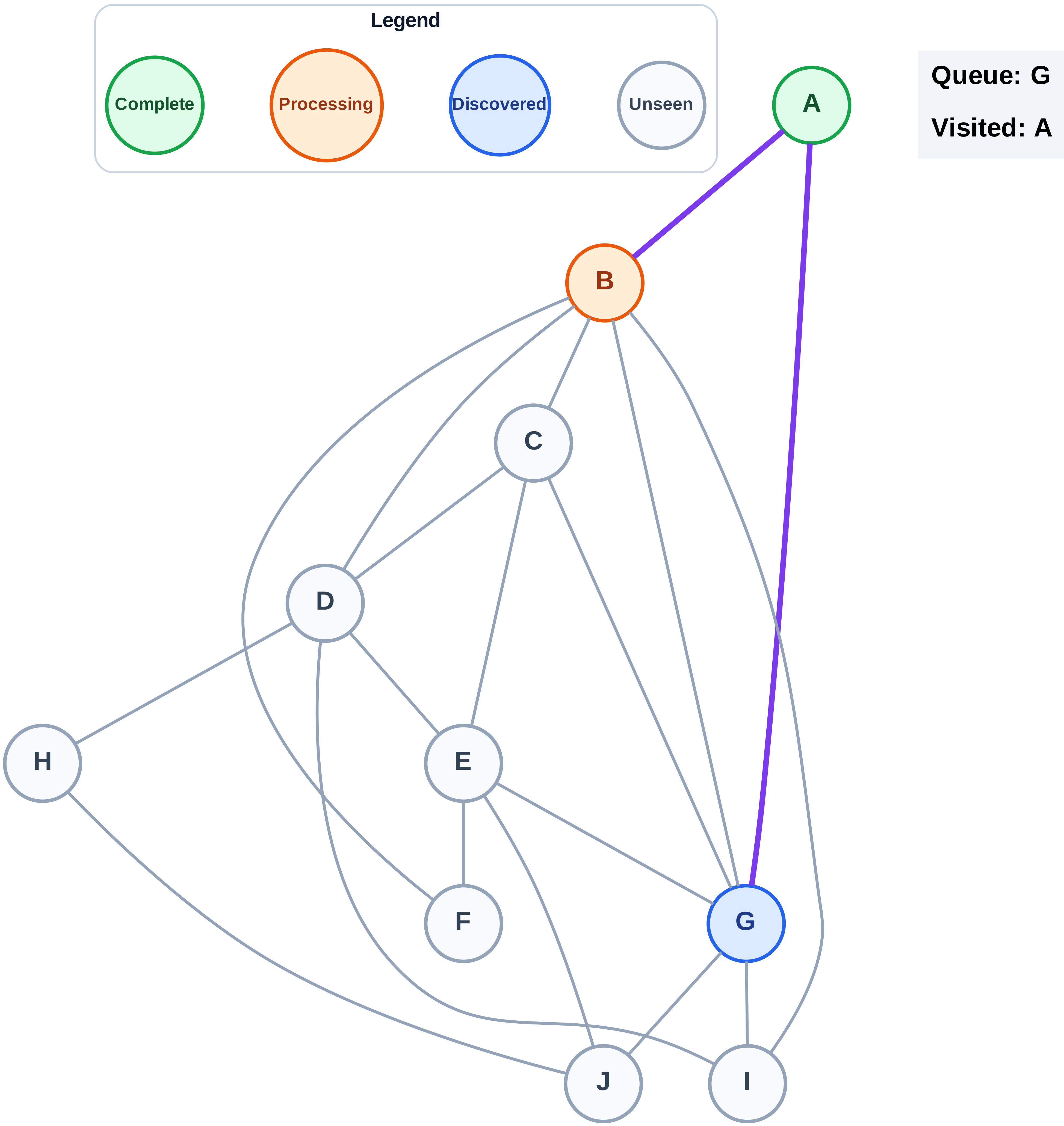
Queue: (empty)

Visited: (none)


BFS Traversal
Step 3: Discover B, G from A
Legend
Complete
Processing
Discovered
Unseen
Queue: B → G
Visited: A
A
B
C
D
E
F
G
H
I
J

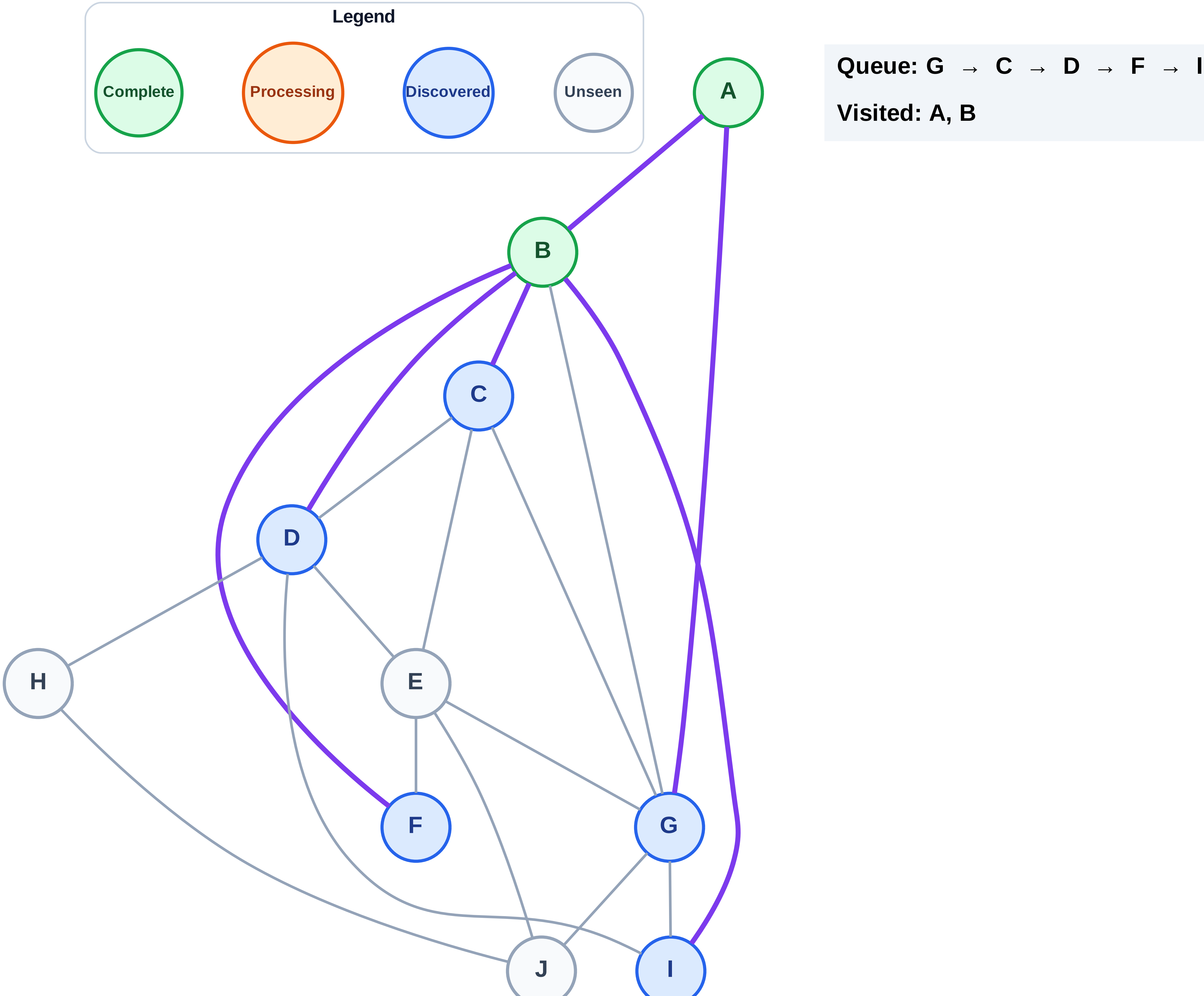
BFS Traversal

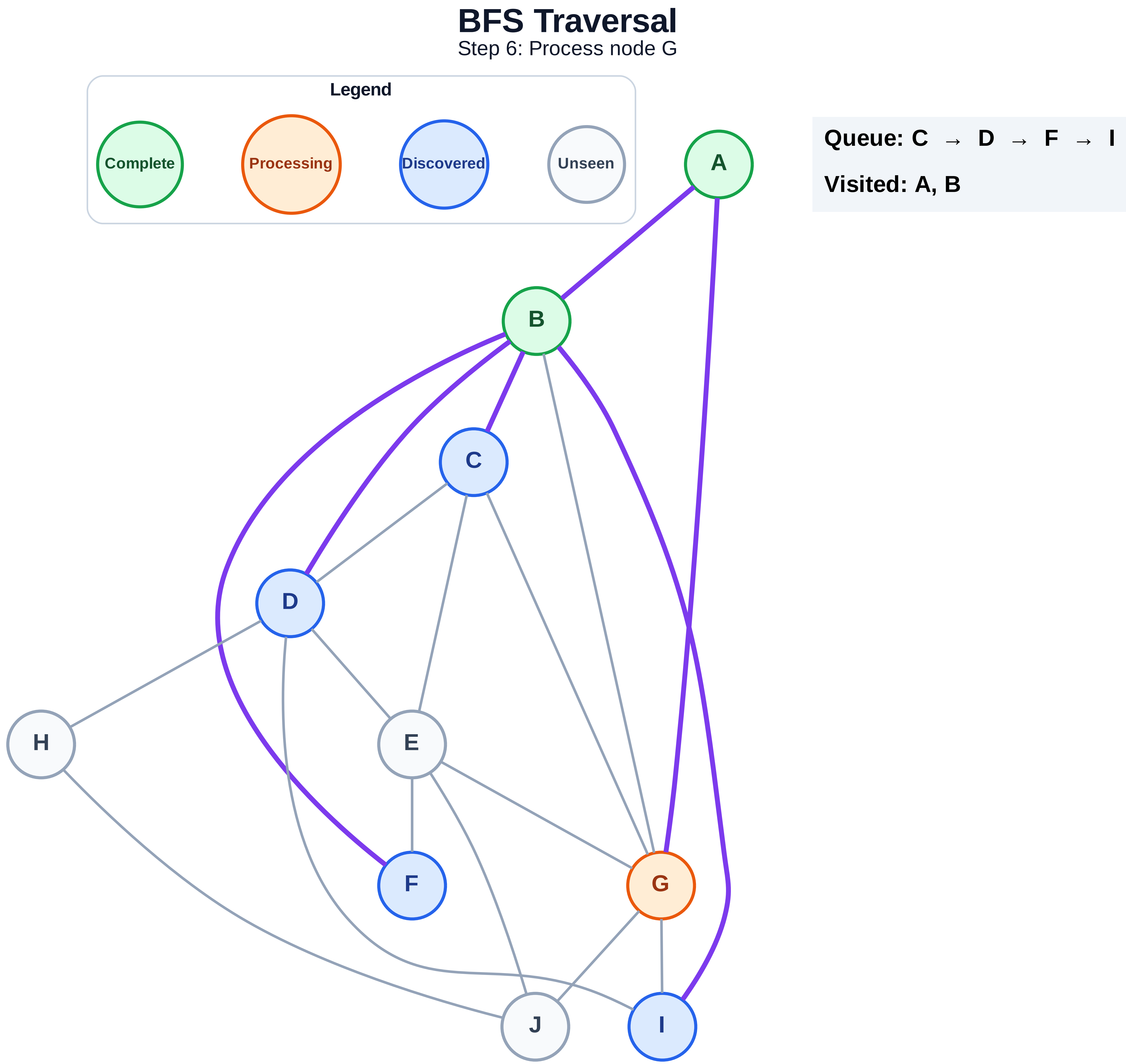
Step 4: Process node B



BFS Traversal

Step 5: Discover C, D, F, I from B

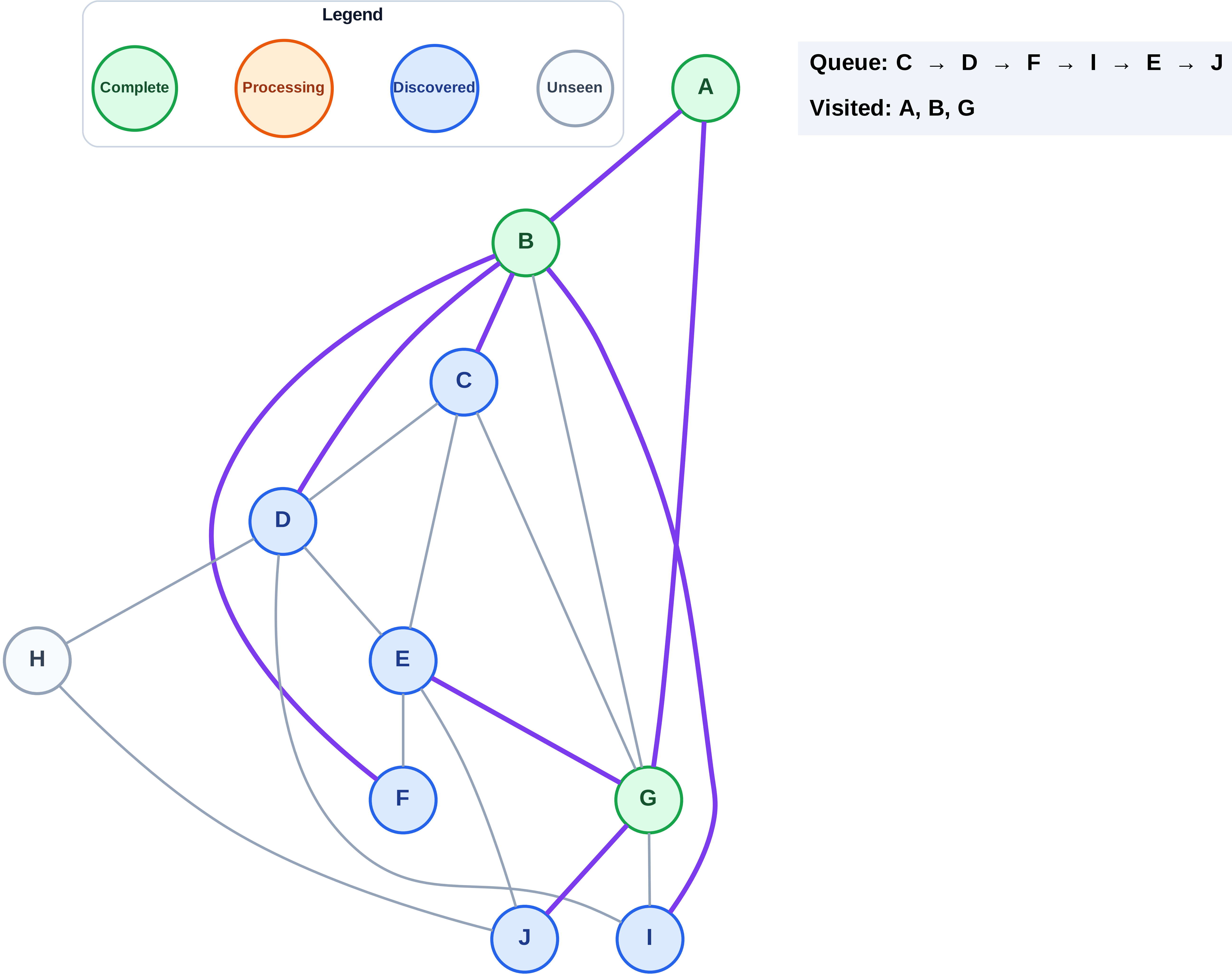


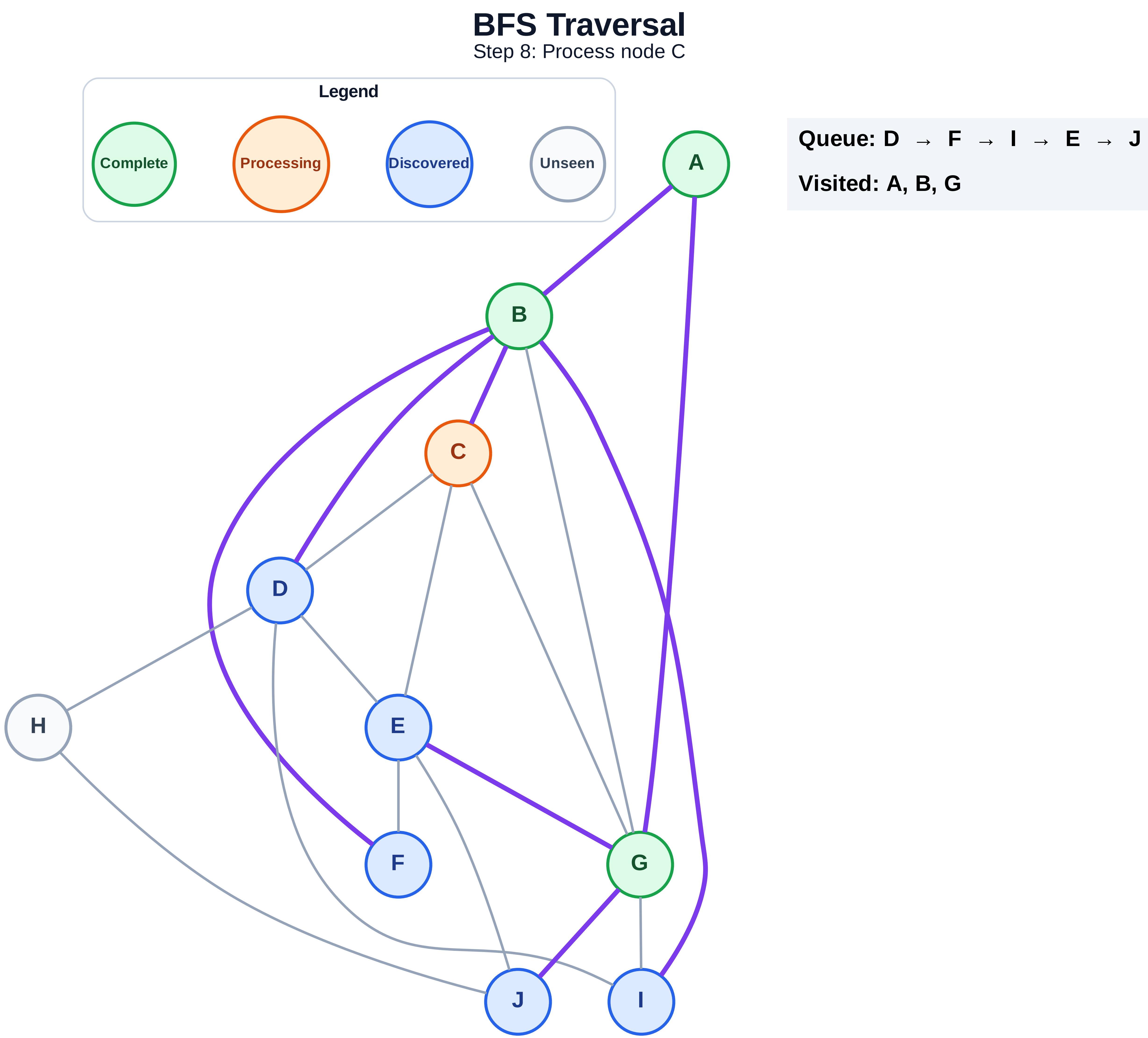


BFS Traversal
Step 6: Process node G
Legend
Complete
Processing
Discovered
Unseen
Queue: C → D → F → I
Visited: A, B
A
B
C
D
E
F
G
H
I
J

BFS Traversal

Step 7: Discover E, J from G

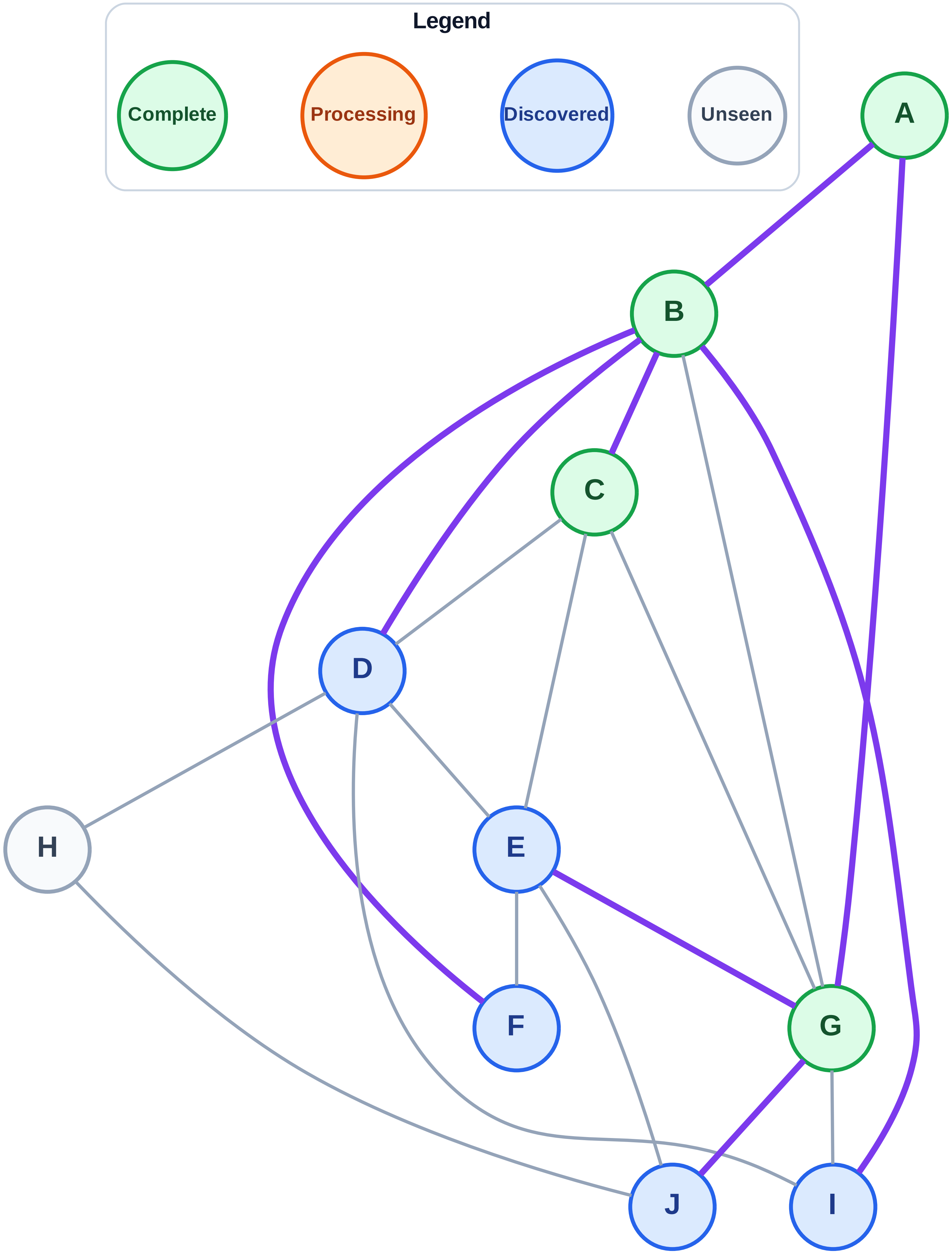




BFS Traversal
Step 8: Process node C
Legend
Complete
Processing
Discovered
Unseen
Queue: D → F → I → E → J
Visited: A, B, G
A
B
C
D
E
F
G
H
I
J

BFS Traversal

Step 9: No new neighbors from C

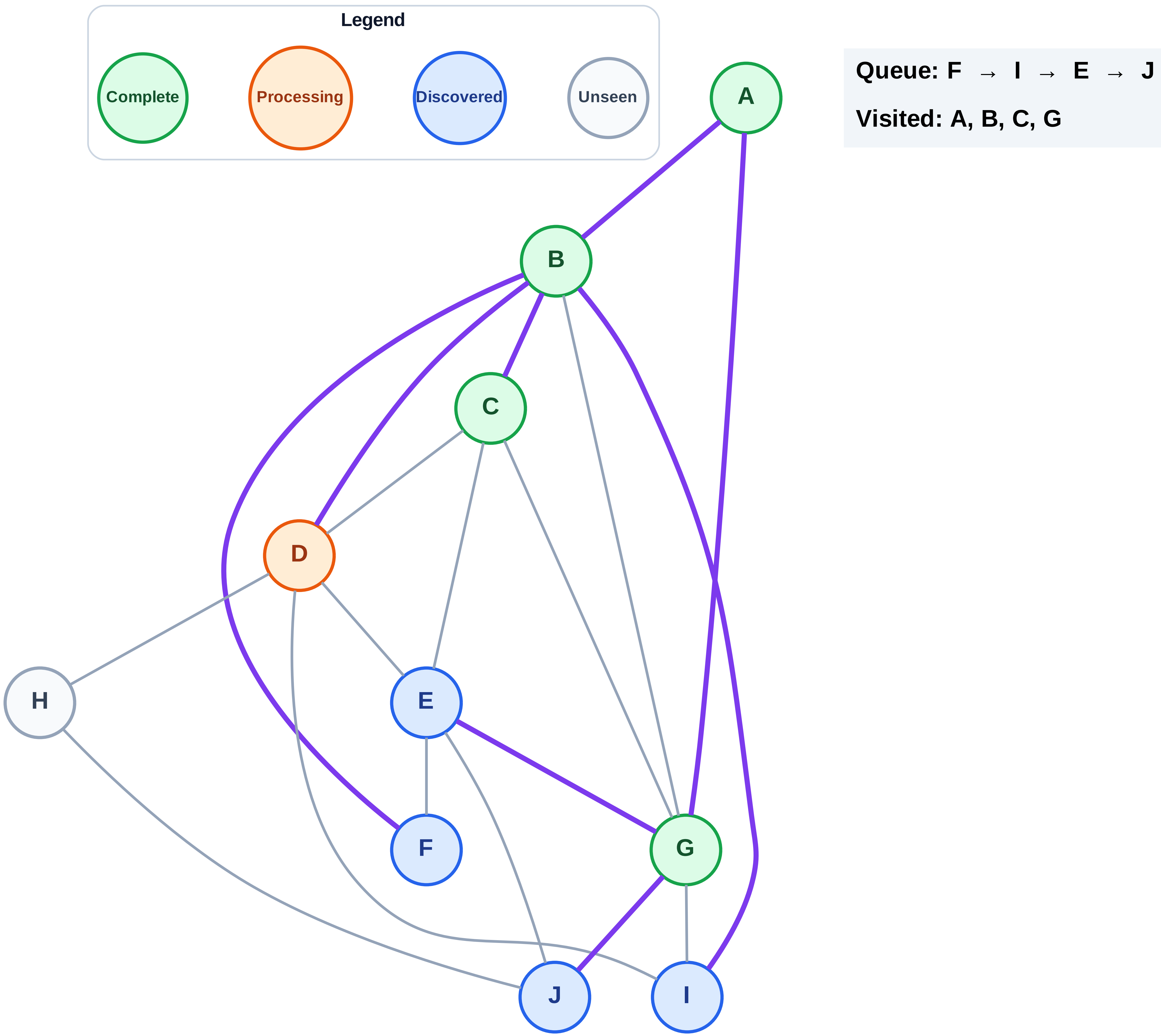


Queue: D → F → I → E → J

Visited: A, B, C, G

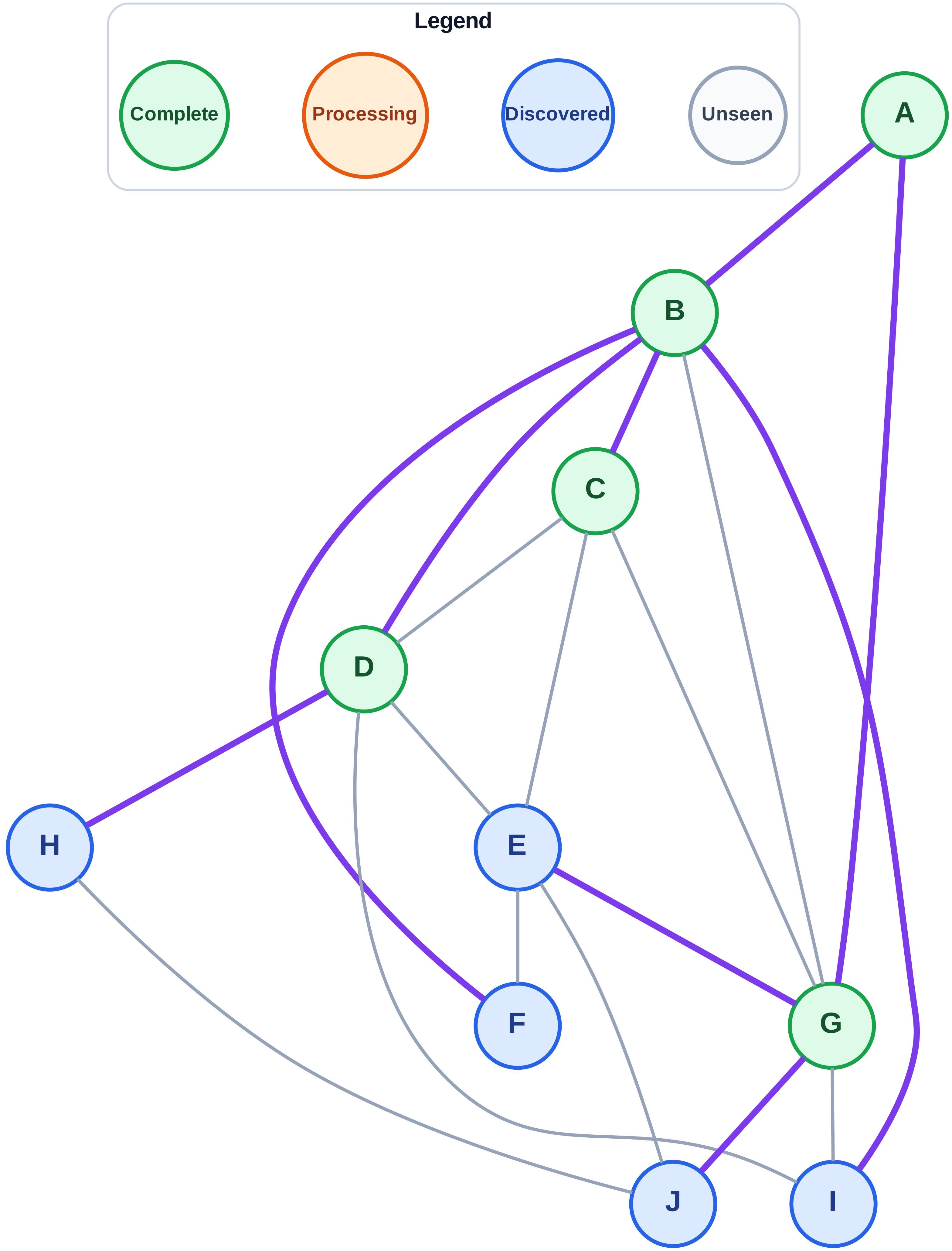
BFS Traversal

Step 10: Process node D



BFS Traversal

Step 11: Discover H from D

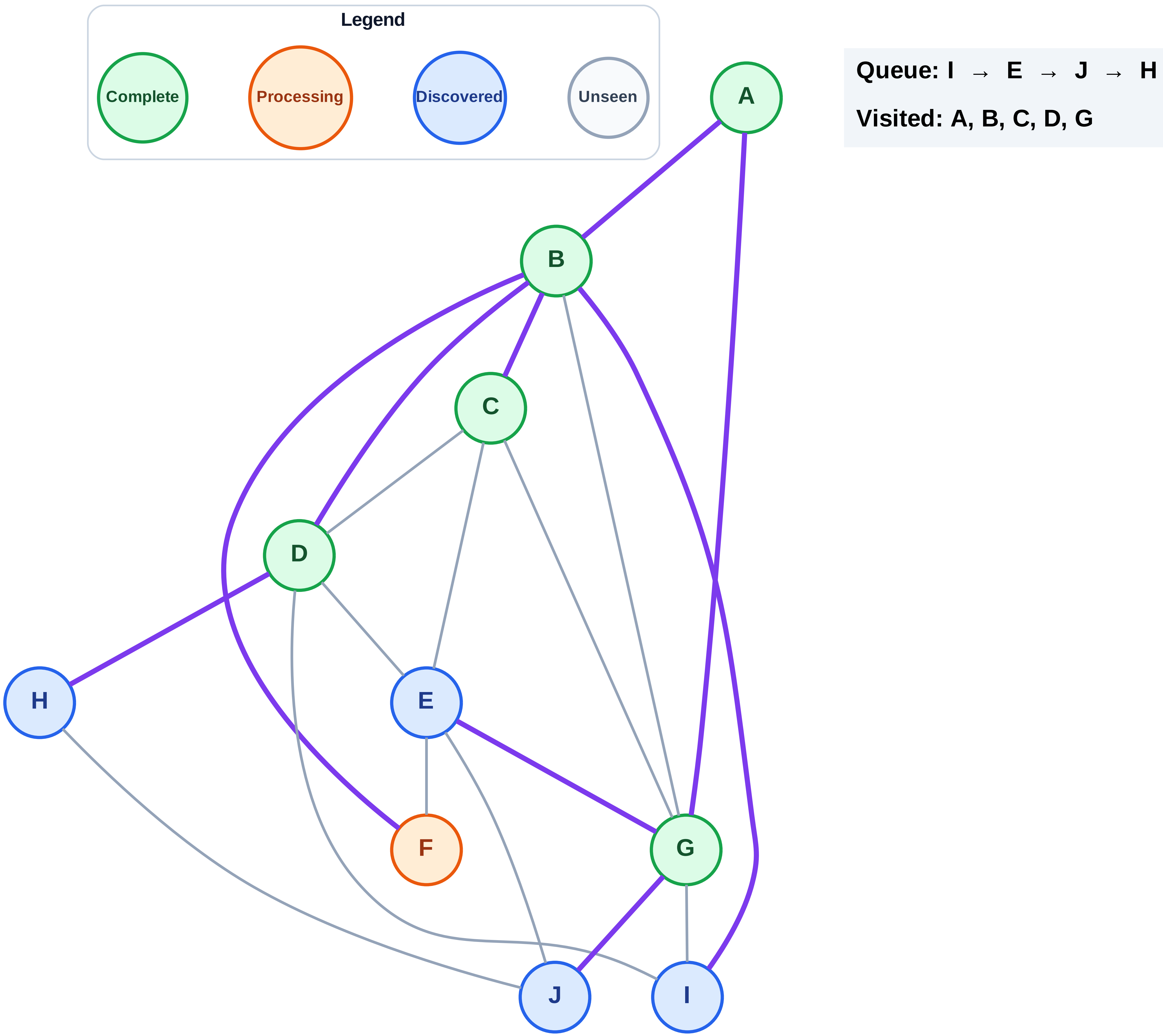


Queue: F → I → E → J → H

Visited: A, B, C, D, G

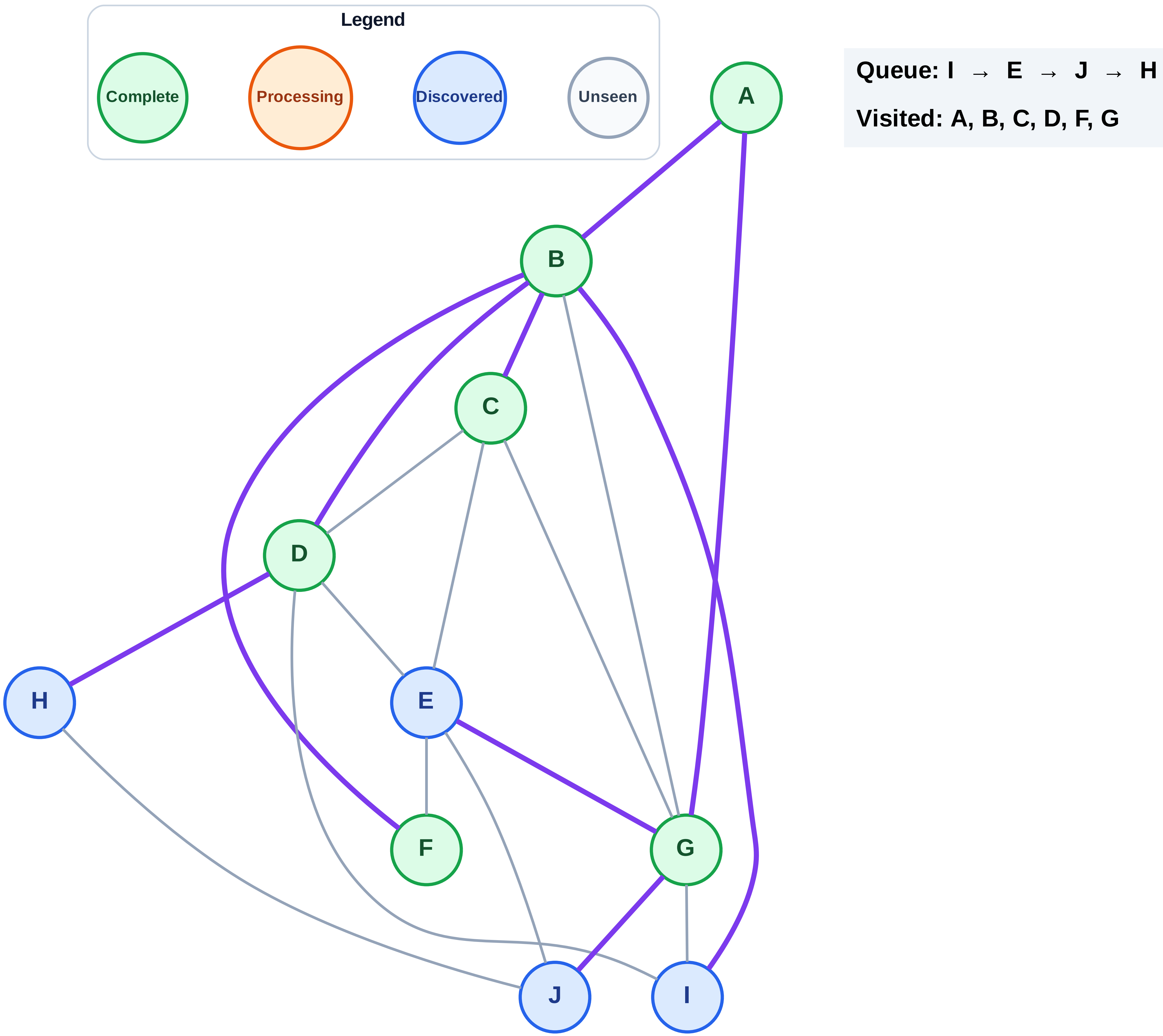
BFS Traversal

Step 12: Process node F



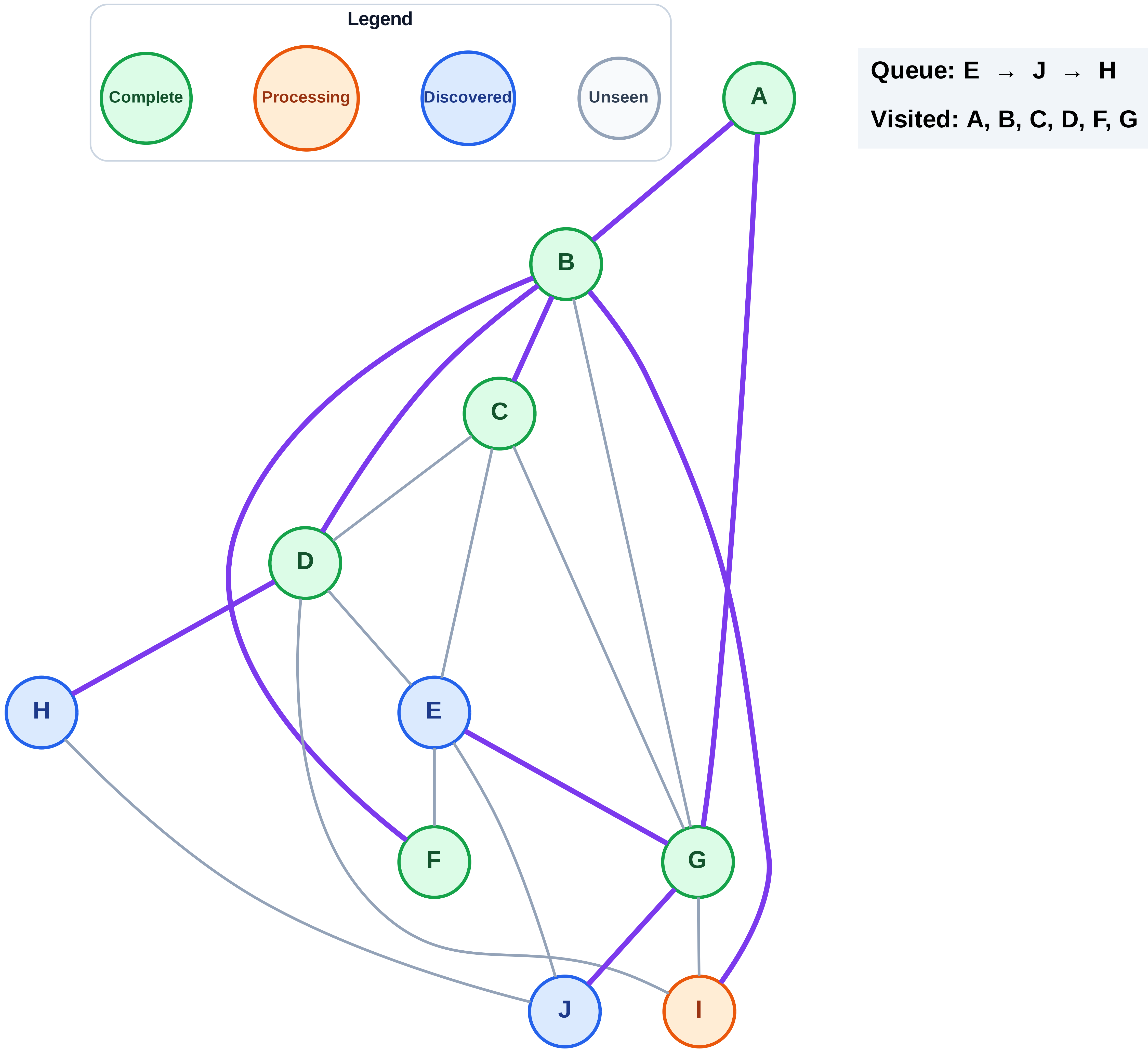
BFS Traversal

Step 13: No new neighbors from F



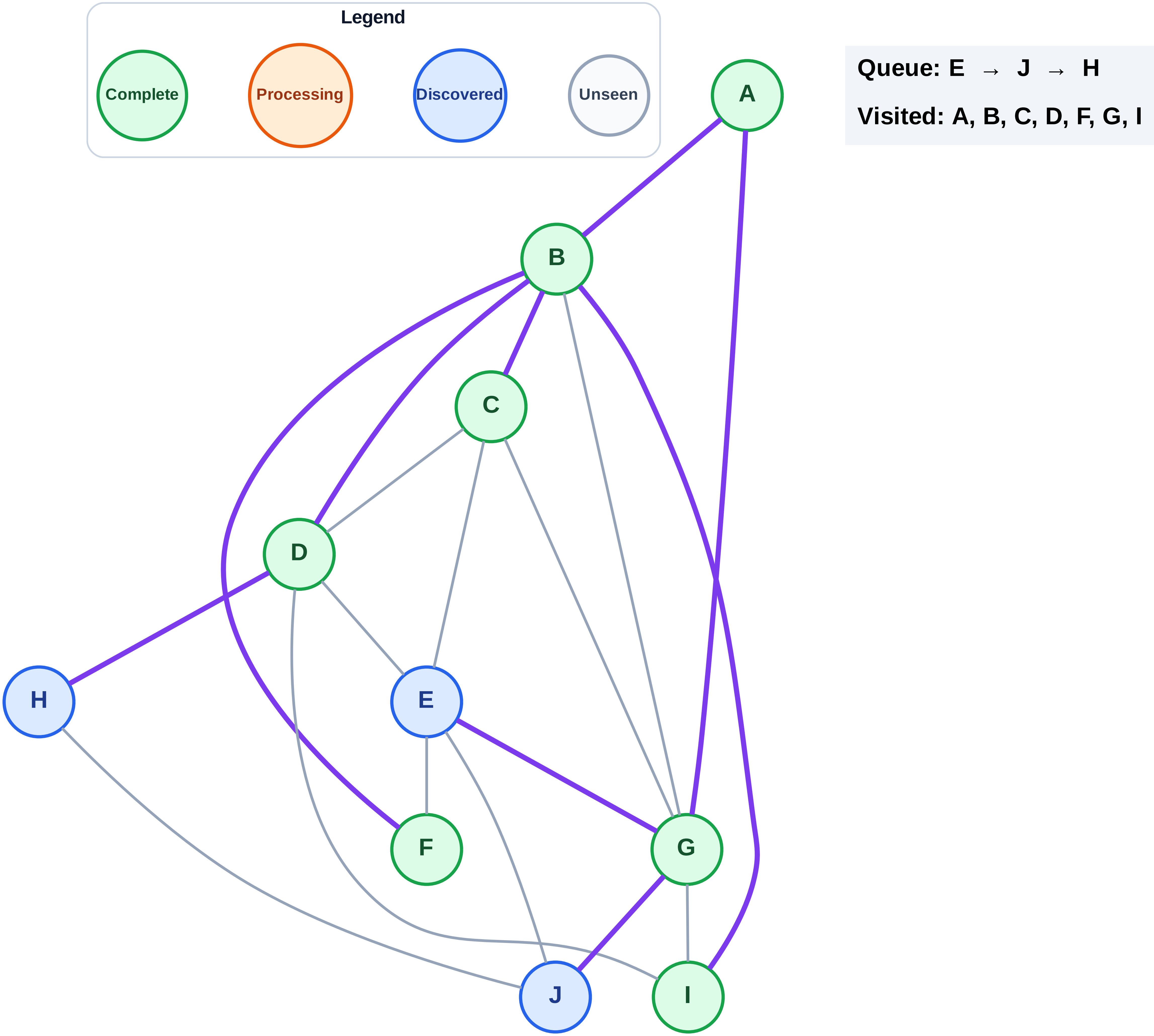
BFS Traversal

Step 14: Process node I



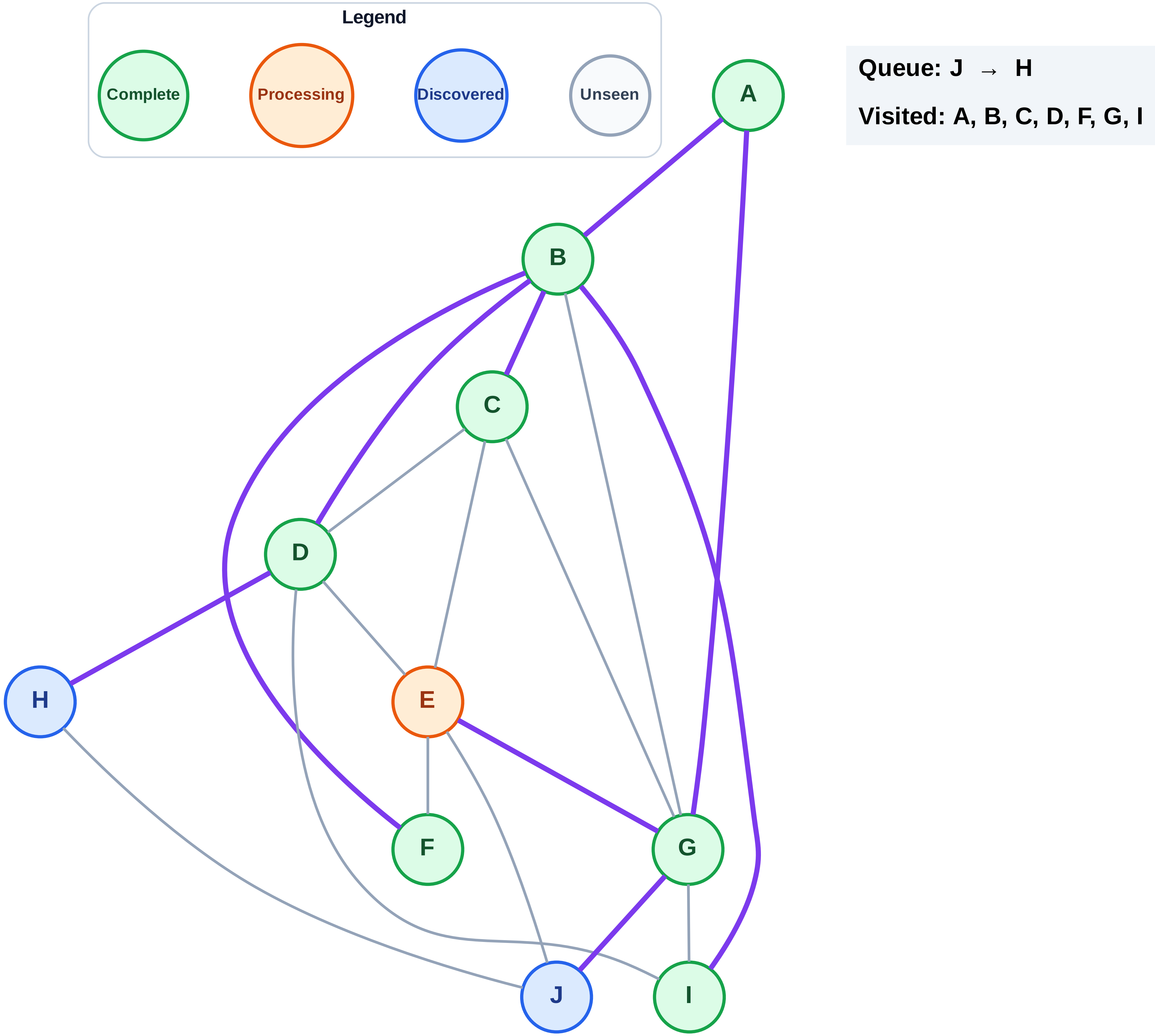
BFS Traversal

Step 15: No new neighbors from I



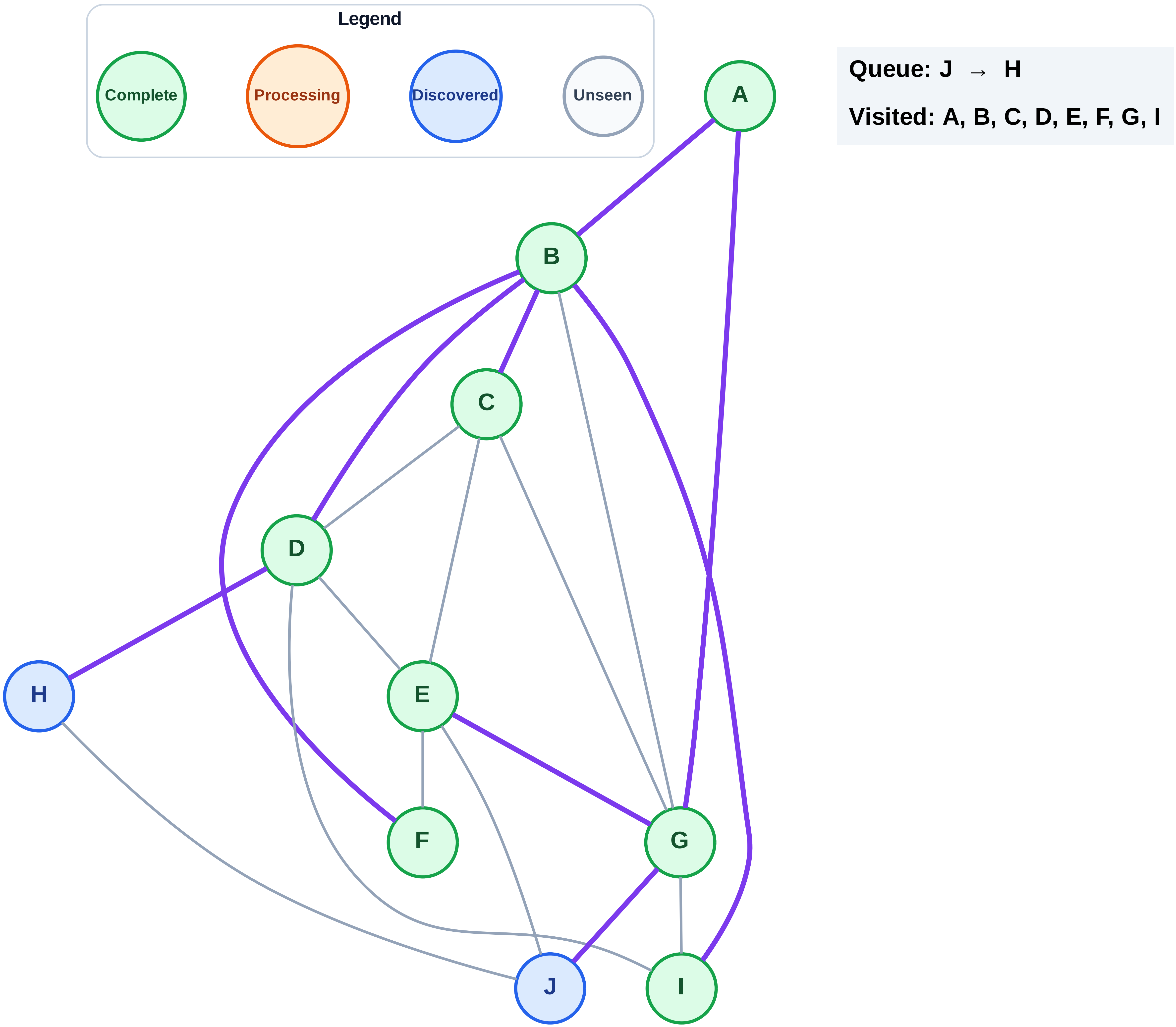
BFS Traversal

Step 16: Process node E



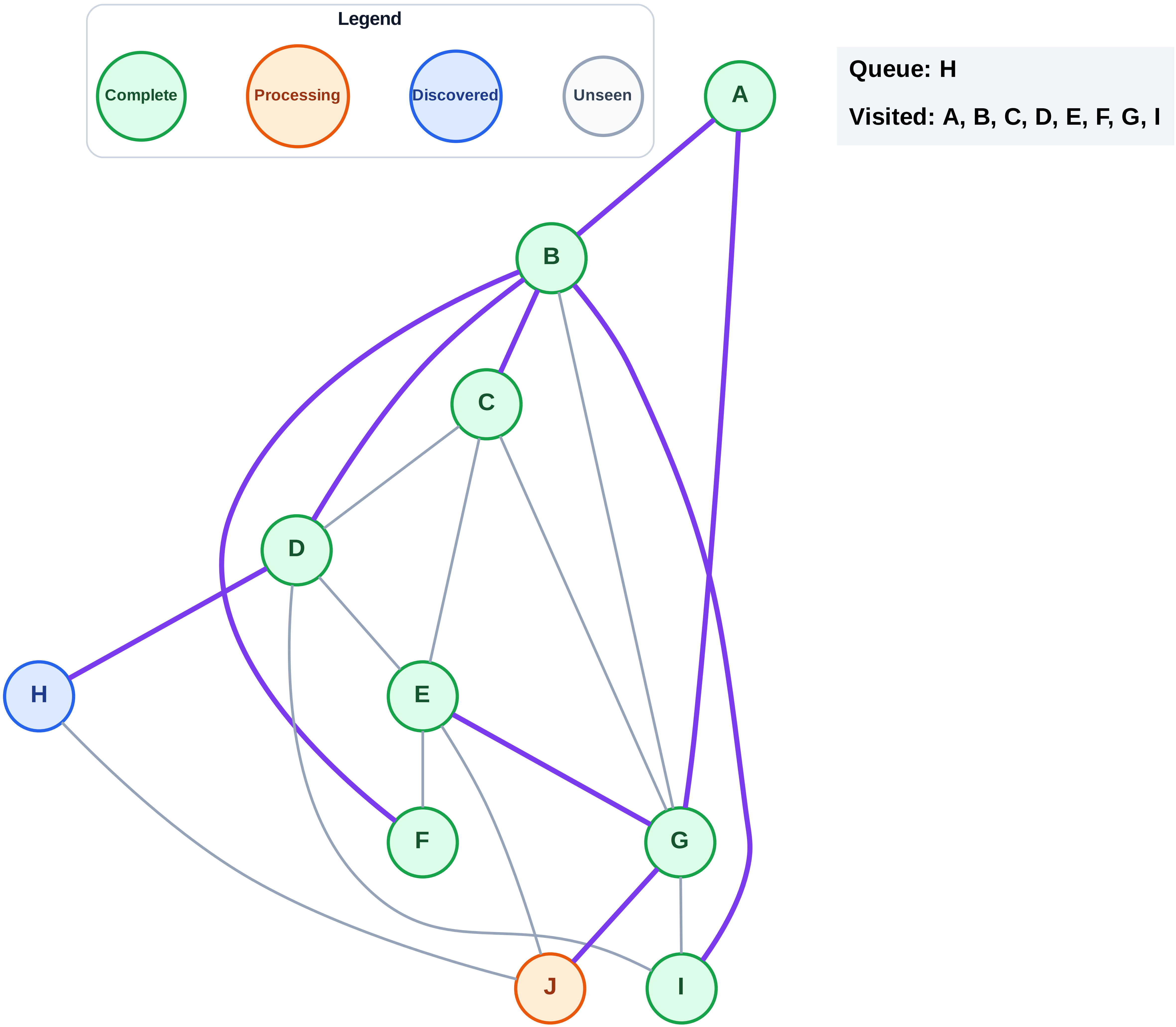
BFS Traversal

Step 17: No new neighbors from E



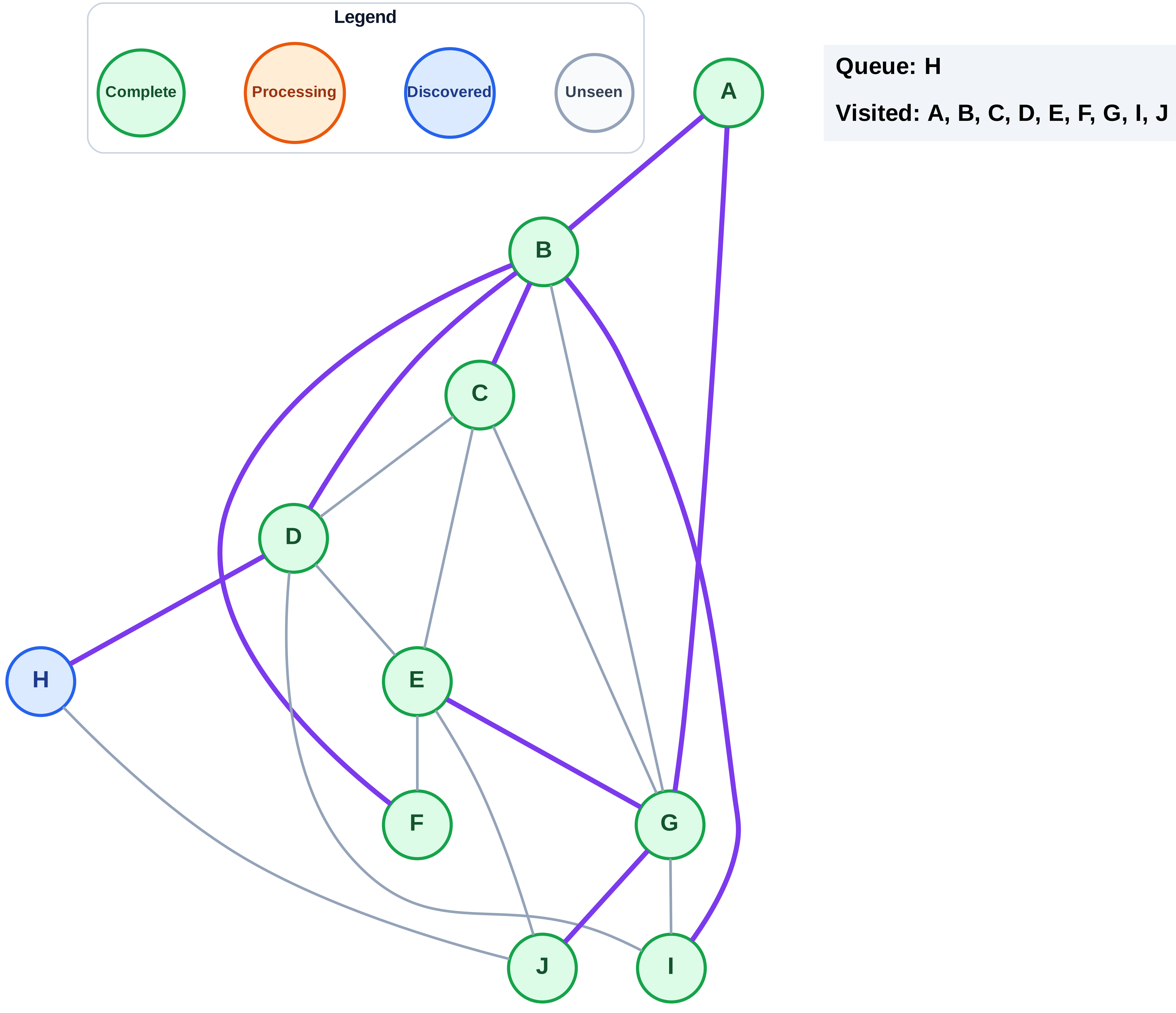
BFS Traversal

Step 18: Process node J



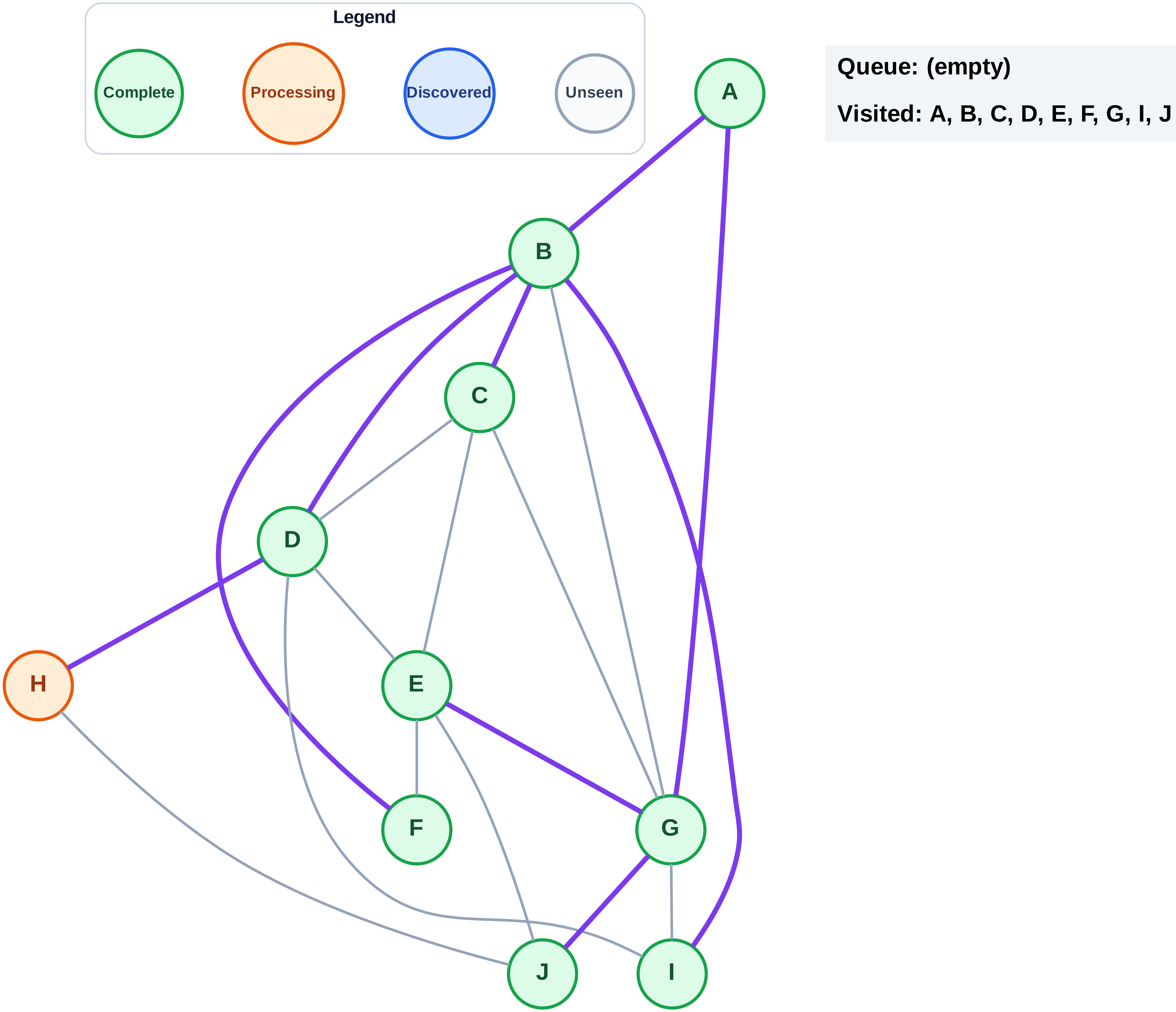
BFS Traversal

Step 19: No new neighbors from J



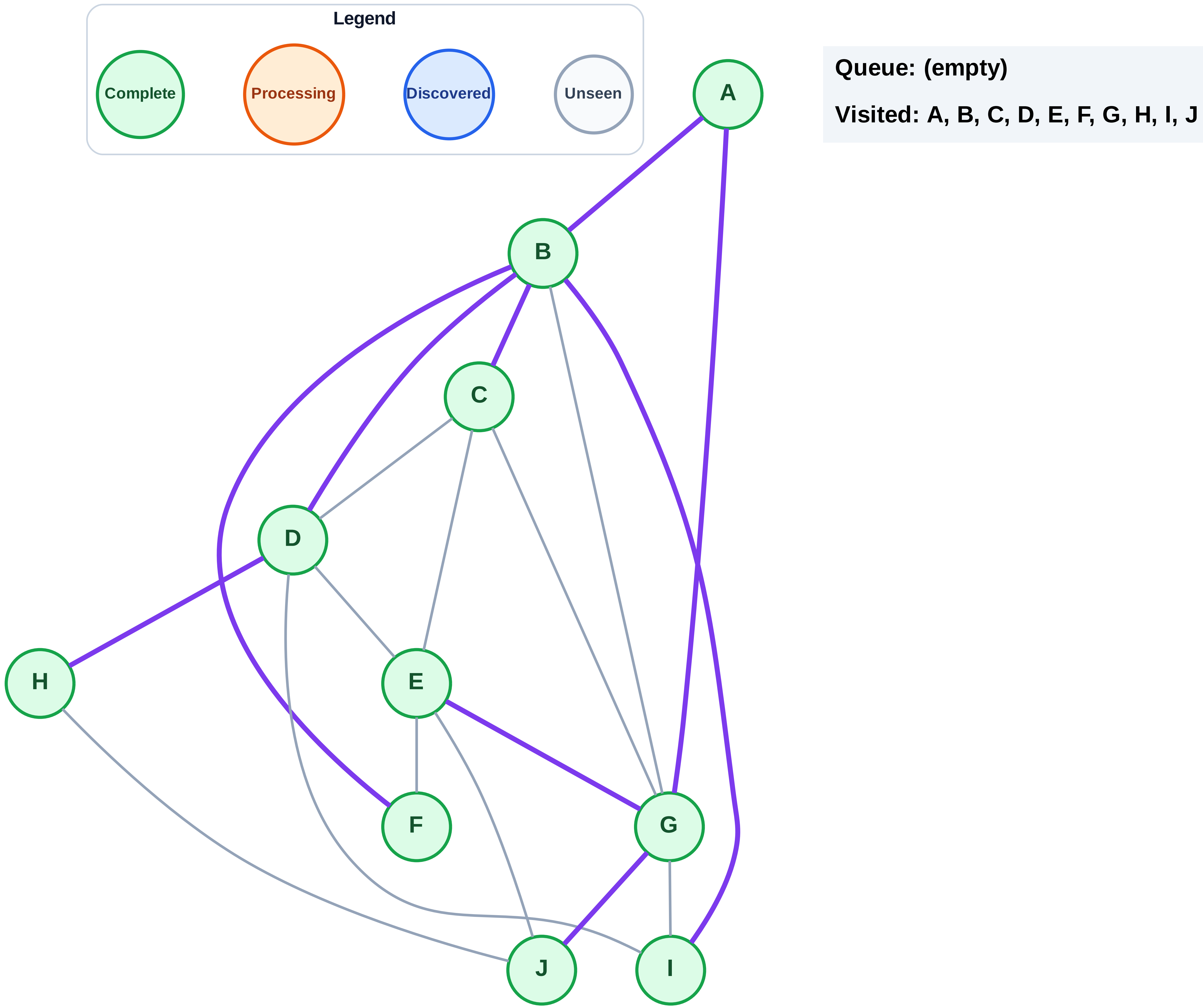
BFS Traversal

Step 20: Process node H



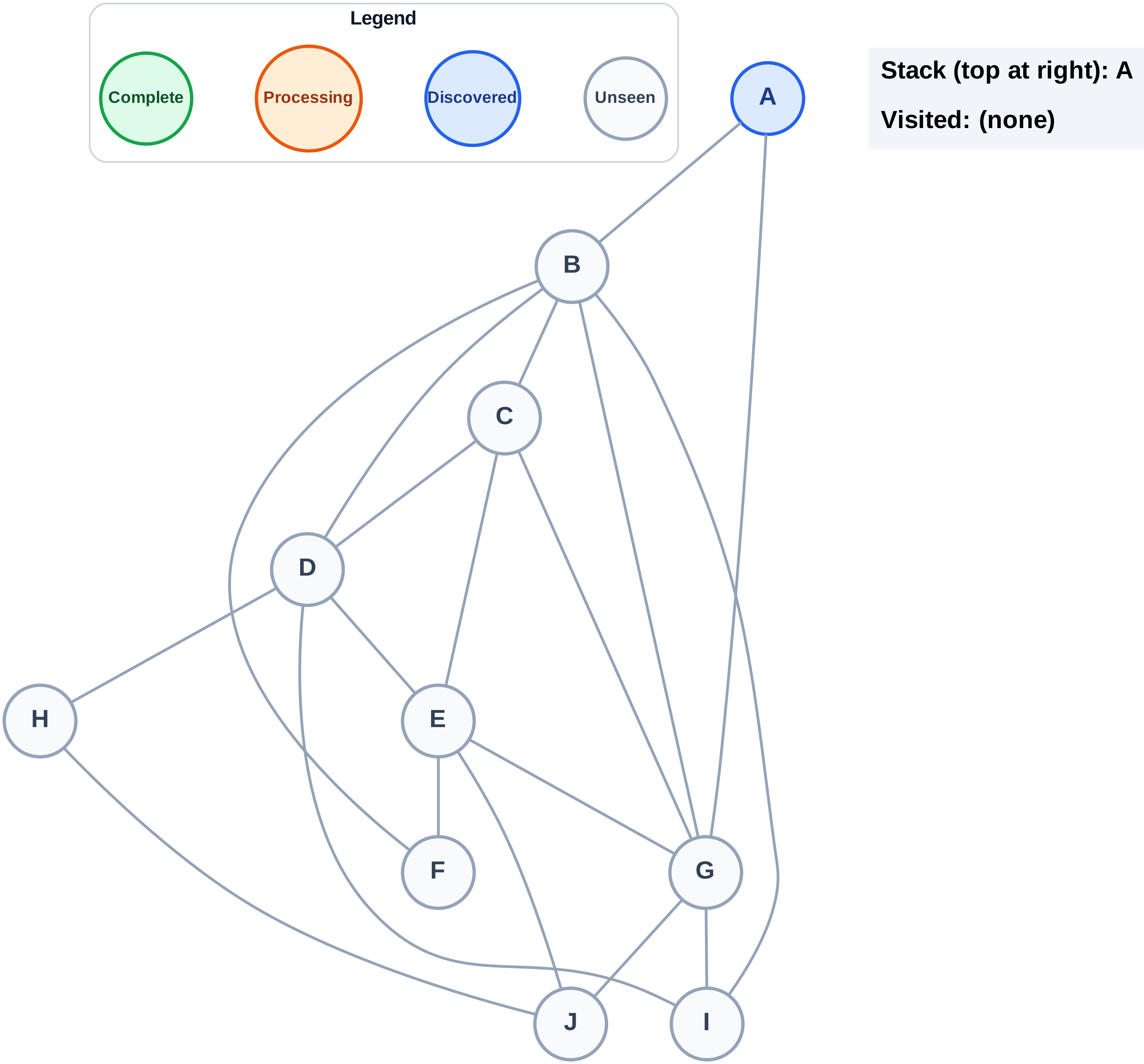
BFS Traversal

Step 21: No new neighbors from H



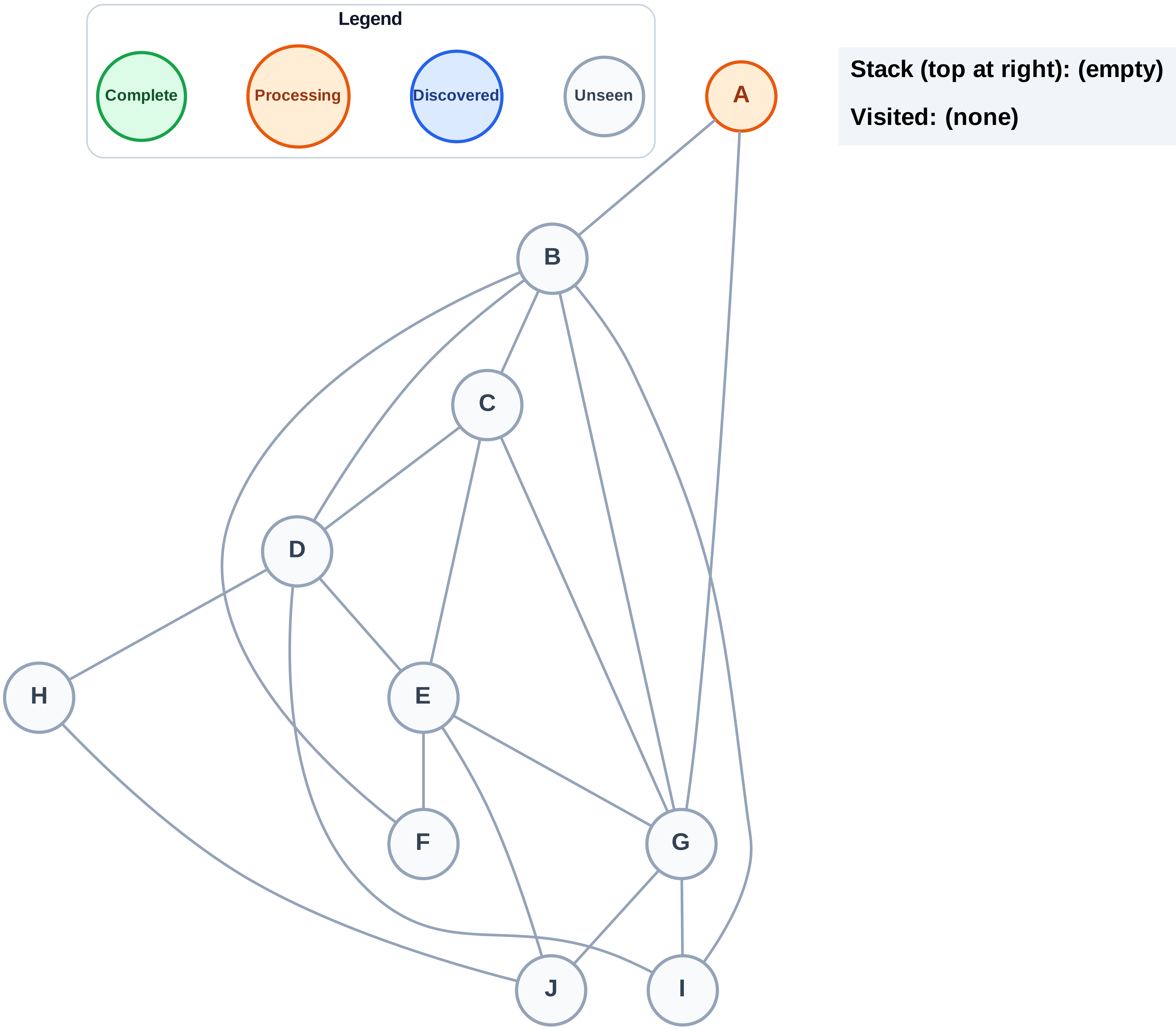
DFS Traversal

Step 1: Start at A



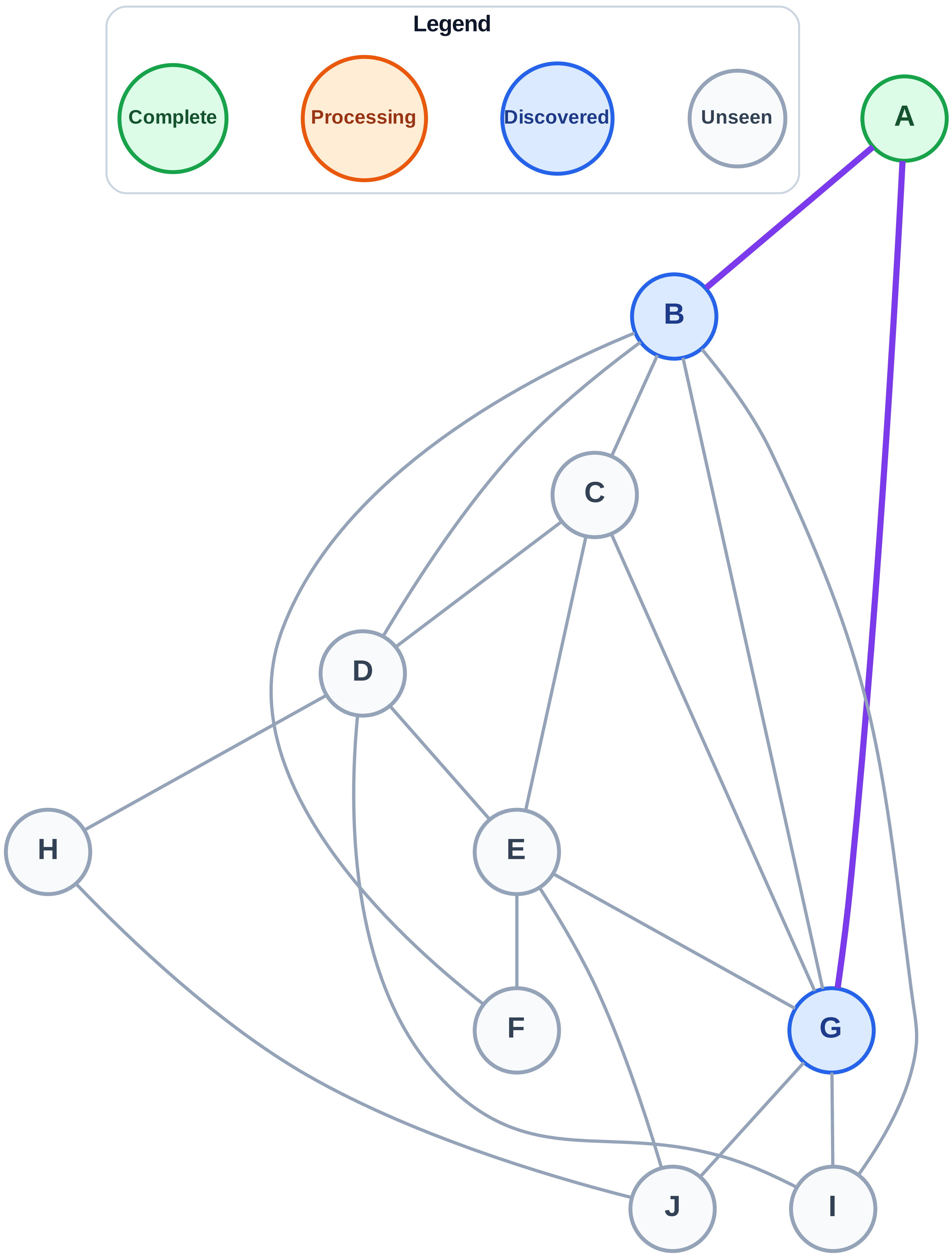
DFS Traversal

Step 2: Process node A



DFS Traversal

Step 3: Discover G, B from A

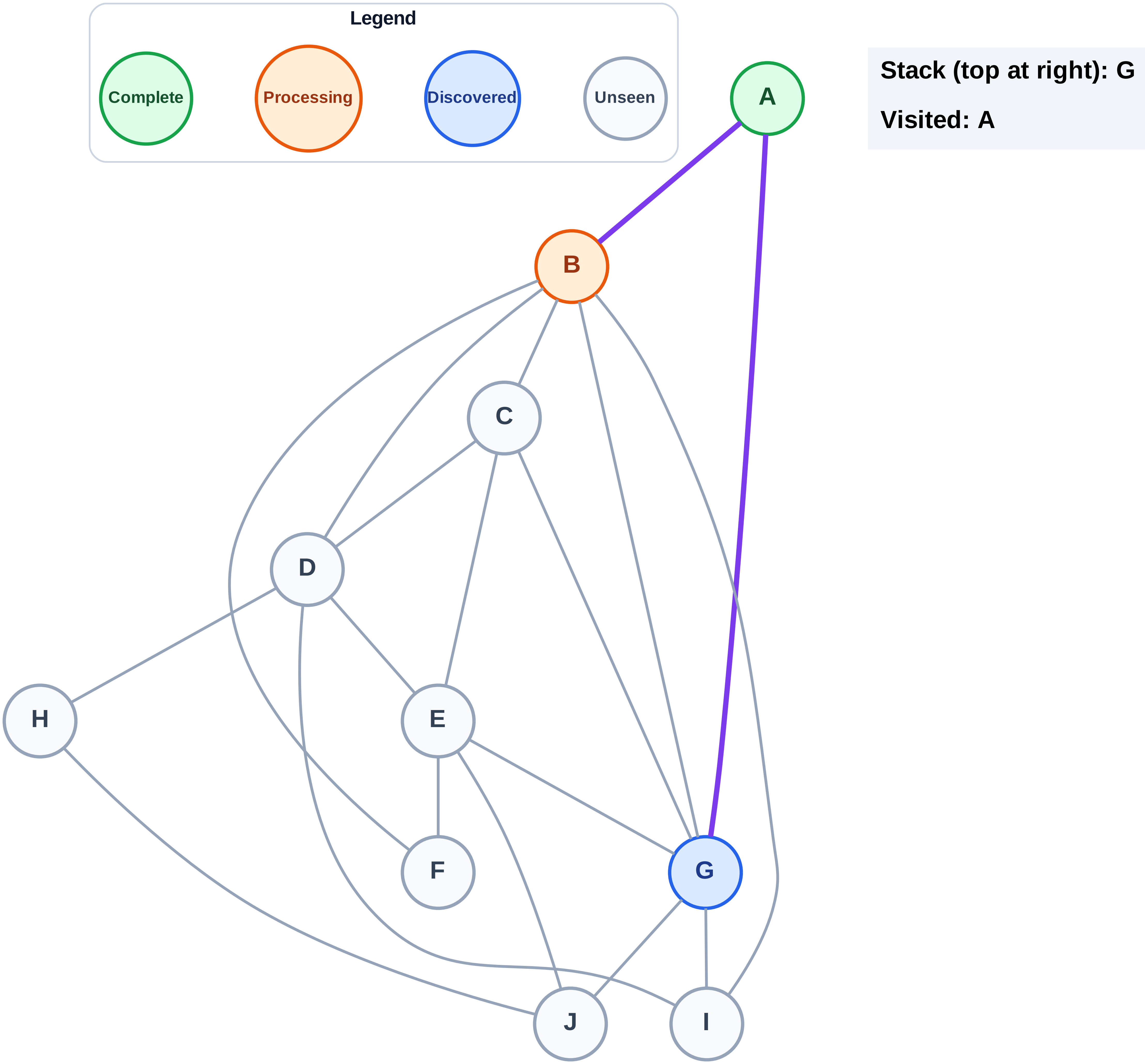


Stack (top at right): G | B

Visited: A

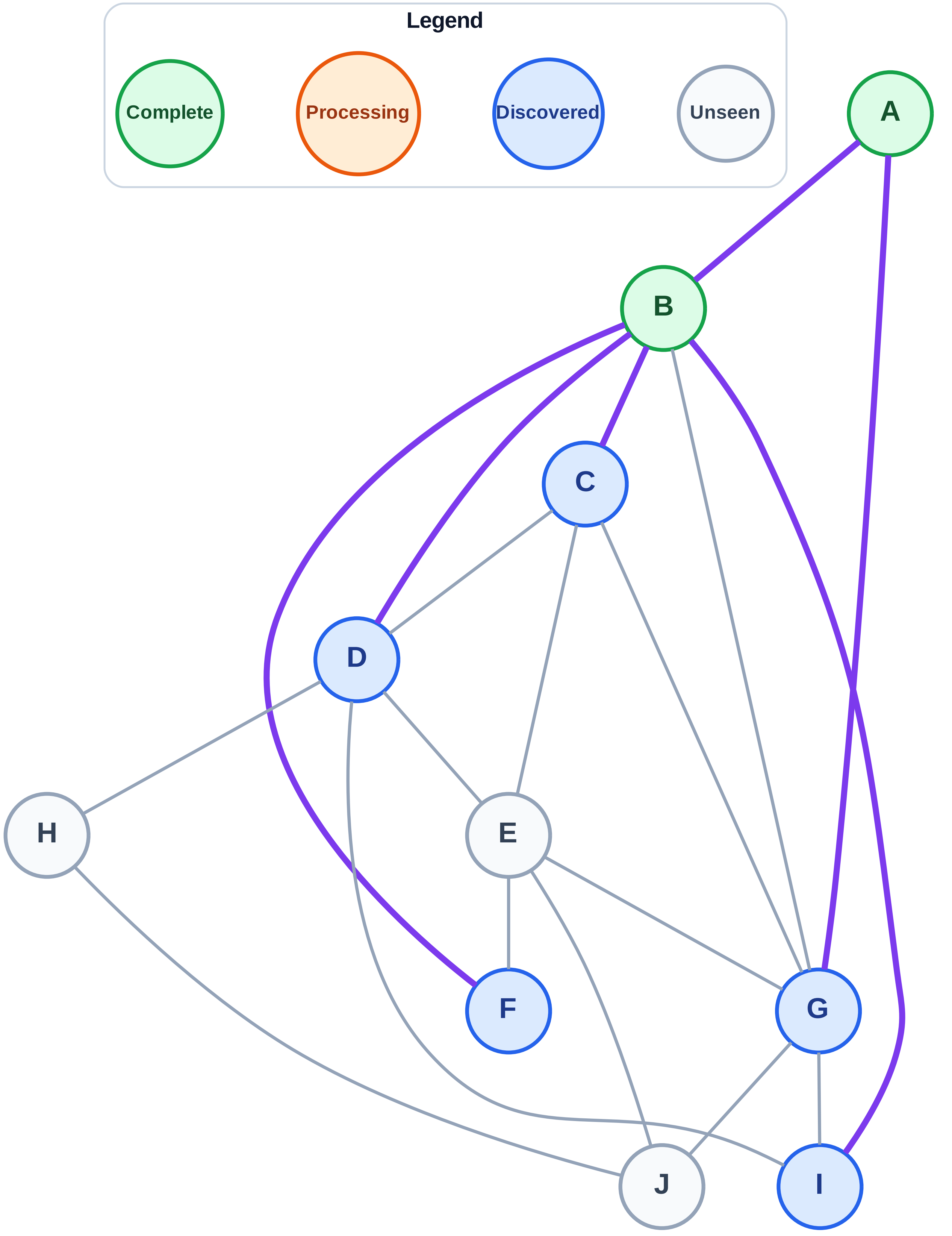
DFS Traversal

Step 4: Process node B



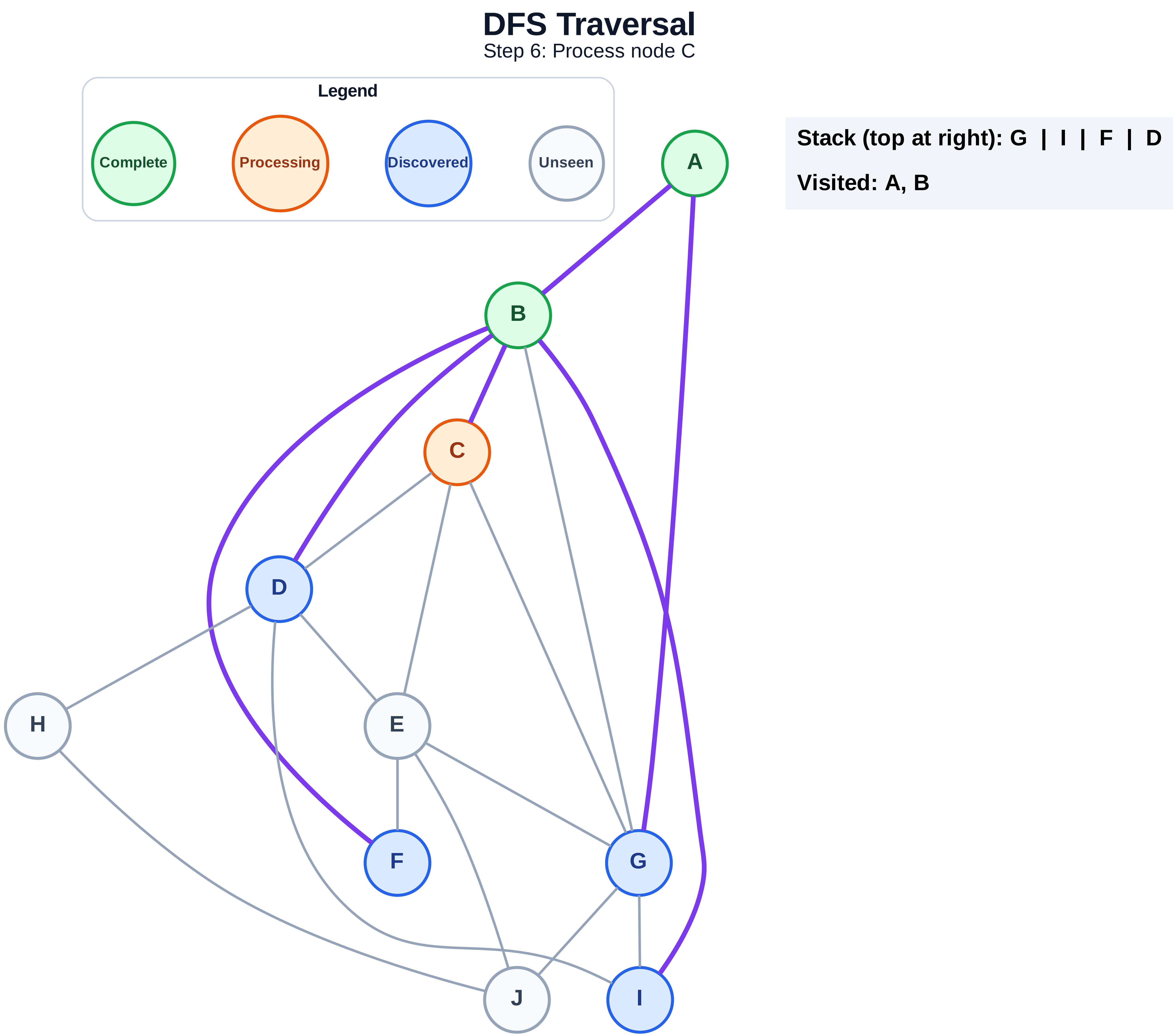
DFS Traversal

Step 5: Discover I, F, D, C from B



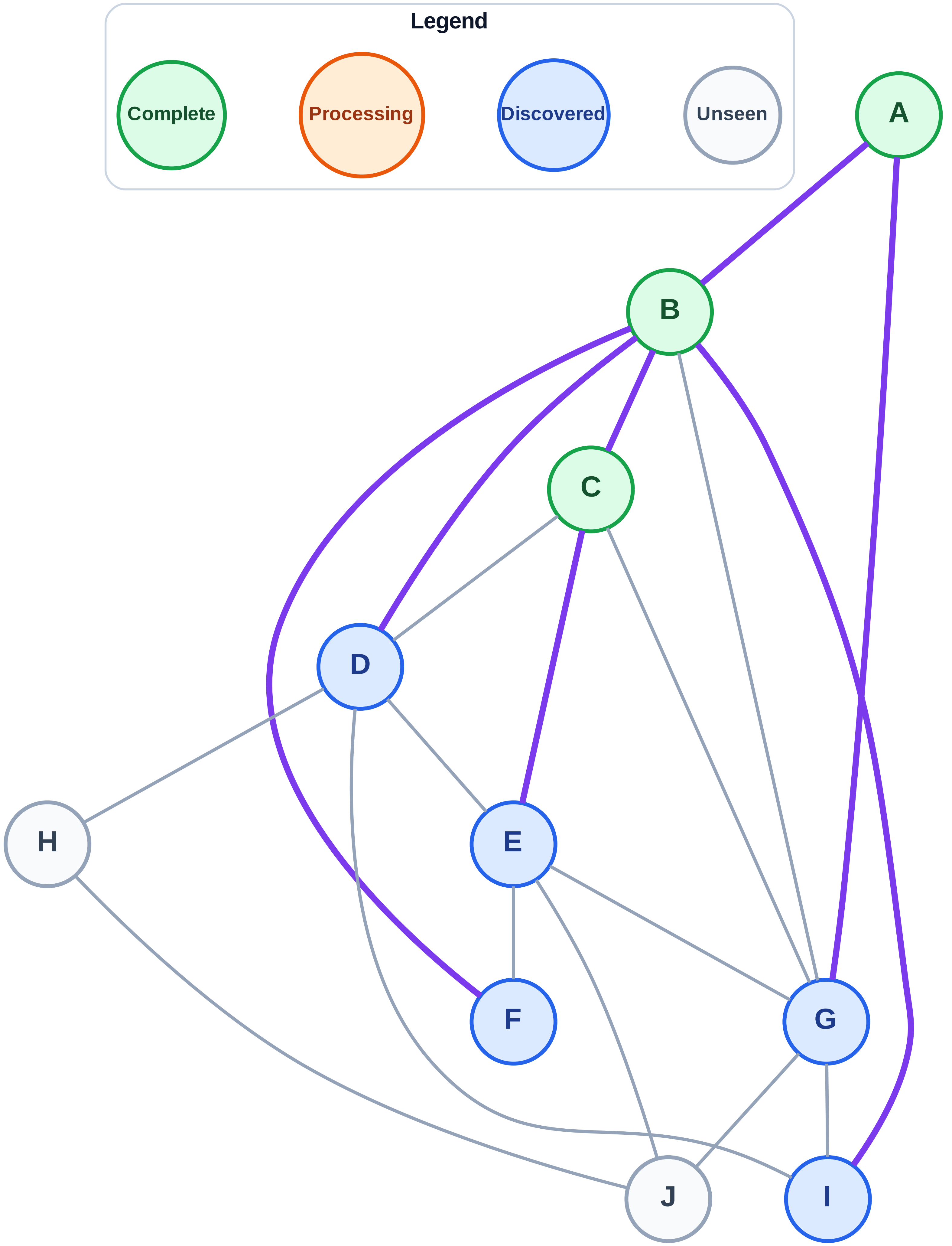
Stack (top at right): G | I | F | D | C

Visited: A, B


DFS Traversal
Step 6: Process node C
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): G | I | F | D
Visited: A, B
A
B
C
D
E
F
G
H
I
J

DFS Traversal

Step 7: Discover E from C

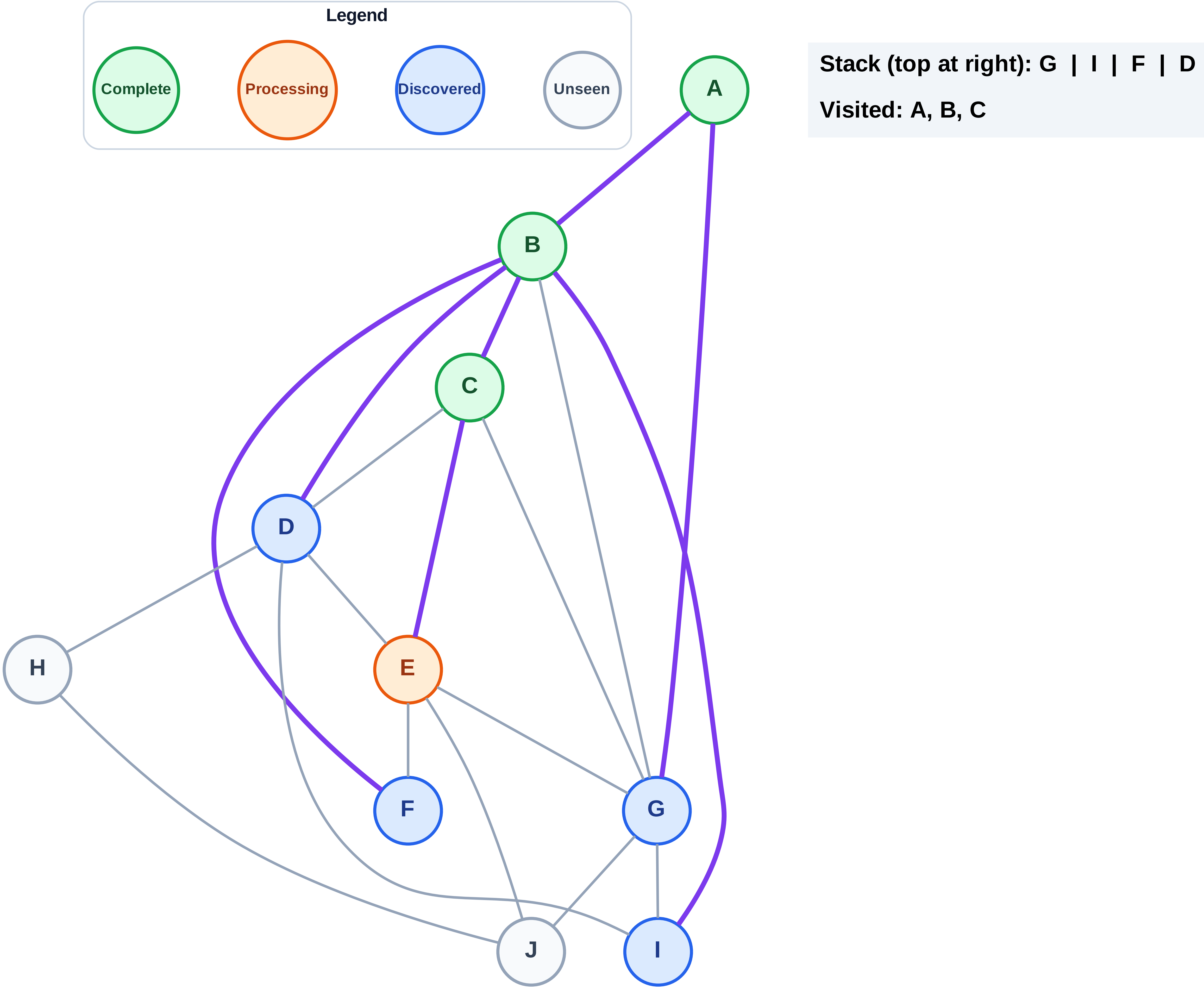


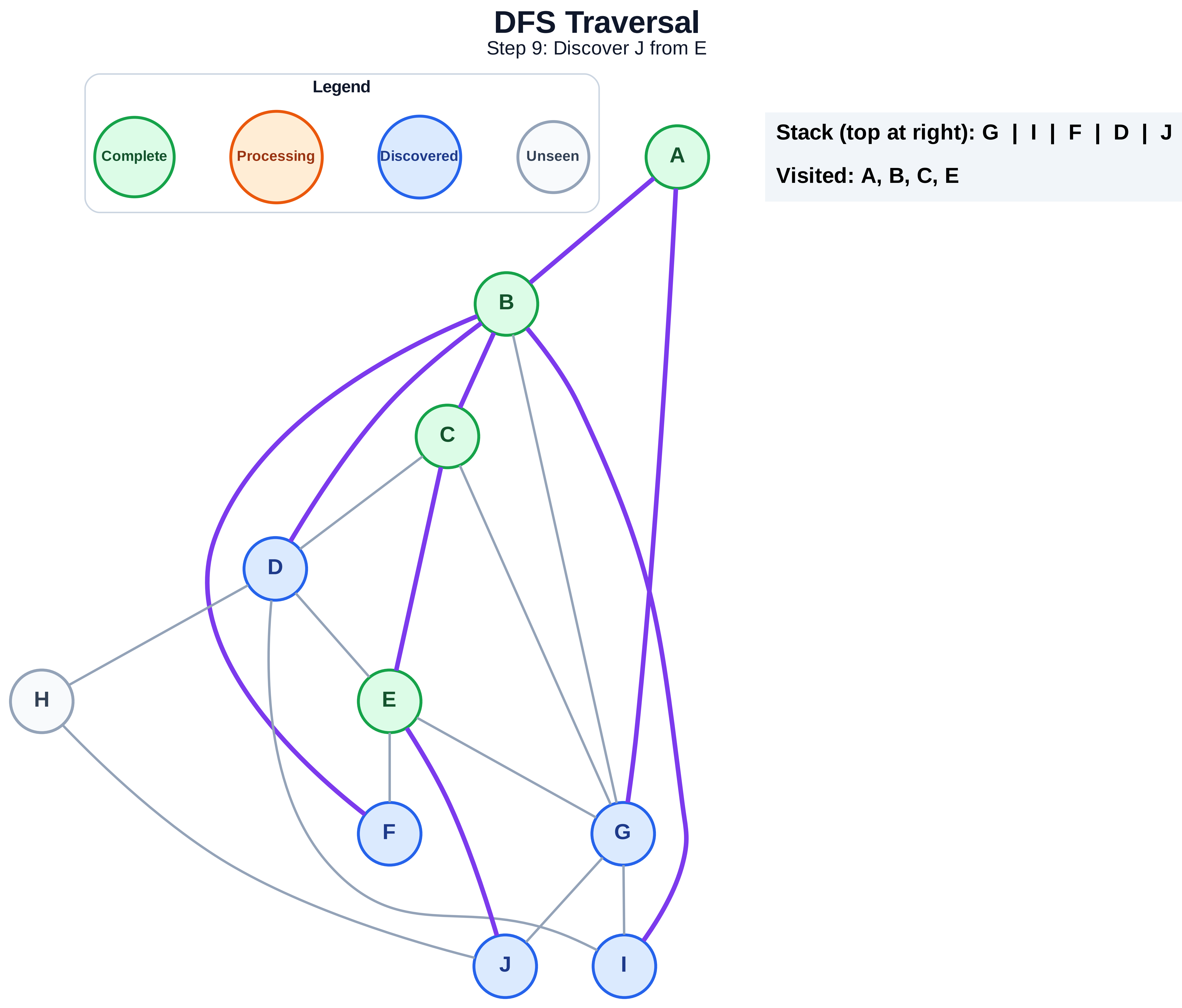
Stack (top at right): G | I | F | D | E

Visited: A, B, C

DFS Traversal

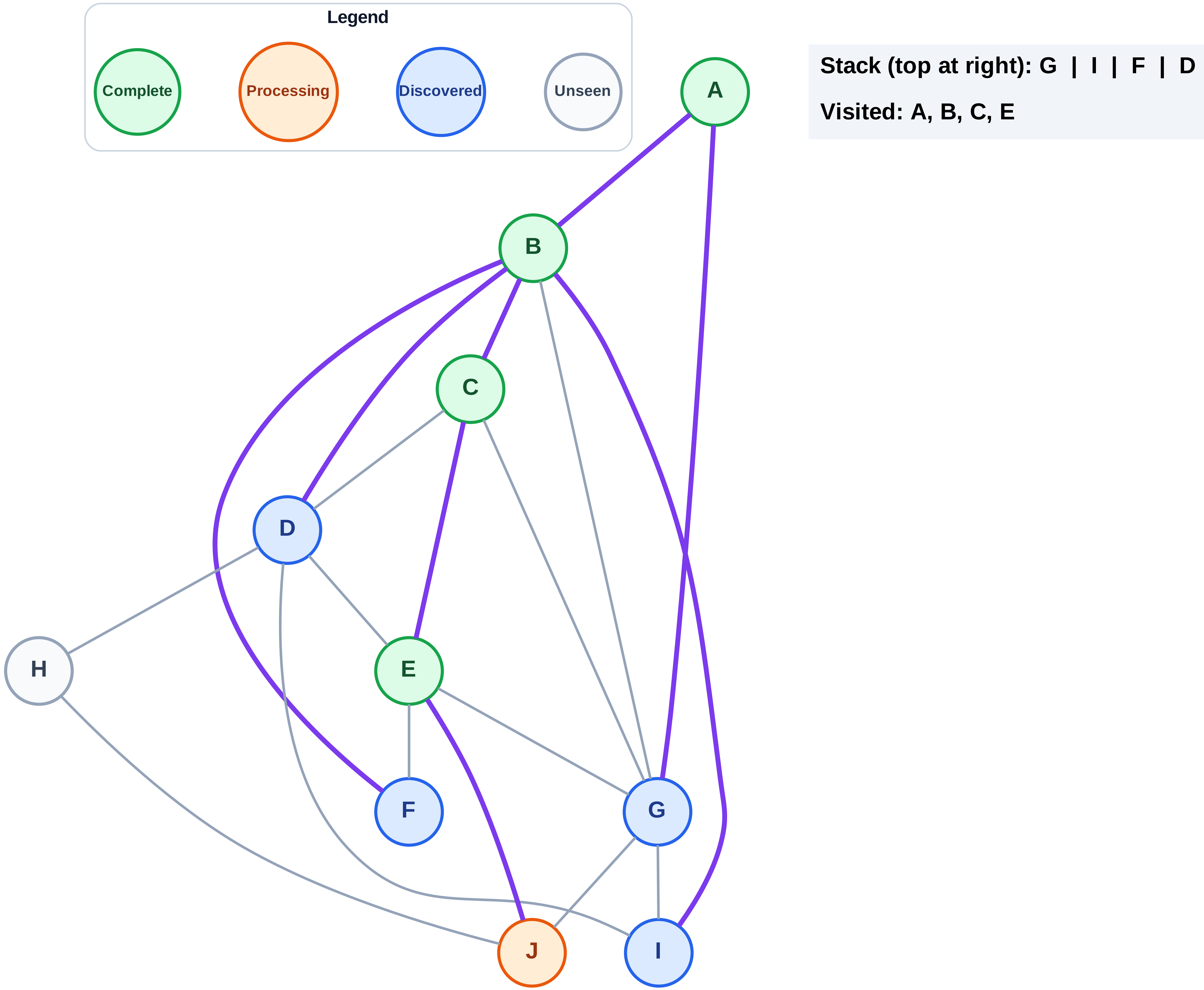
Step 8: Process node E

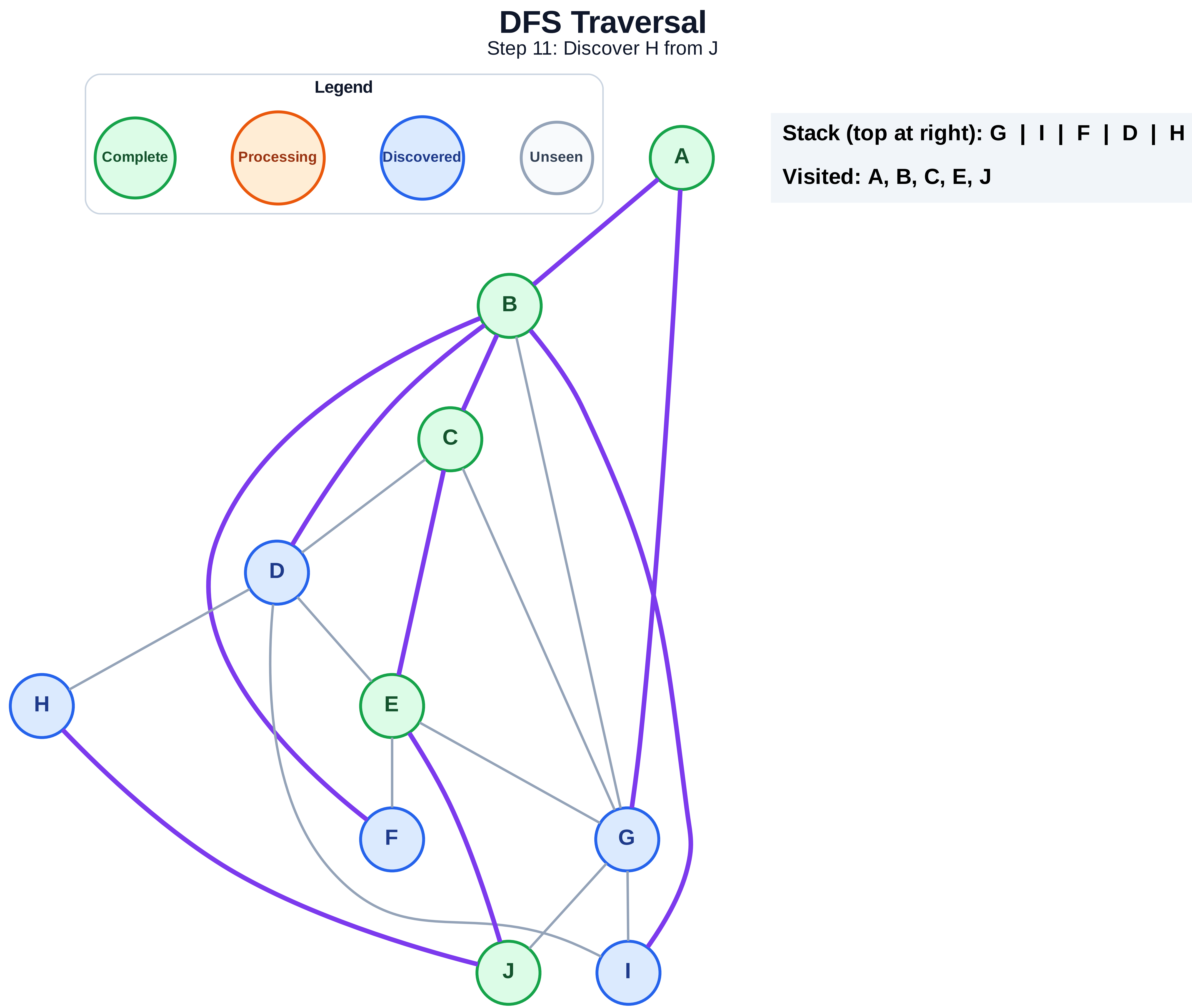



DFS Traversal
Step 9: Discover J from E
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): G | I | F | D | J
Visited: A, B, C, E
A
B
C
D
E
F
G
H
I
J

DFS Traversal

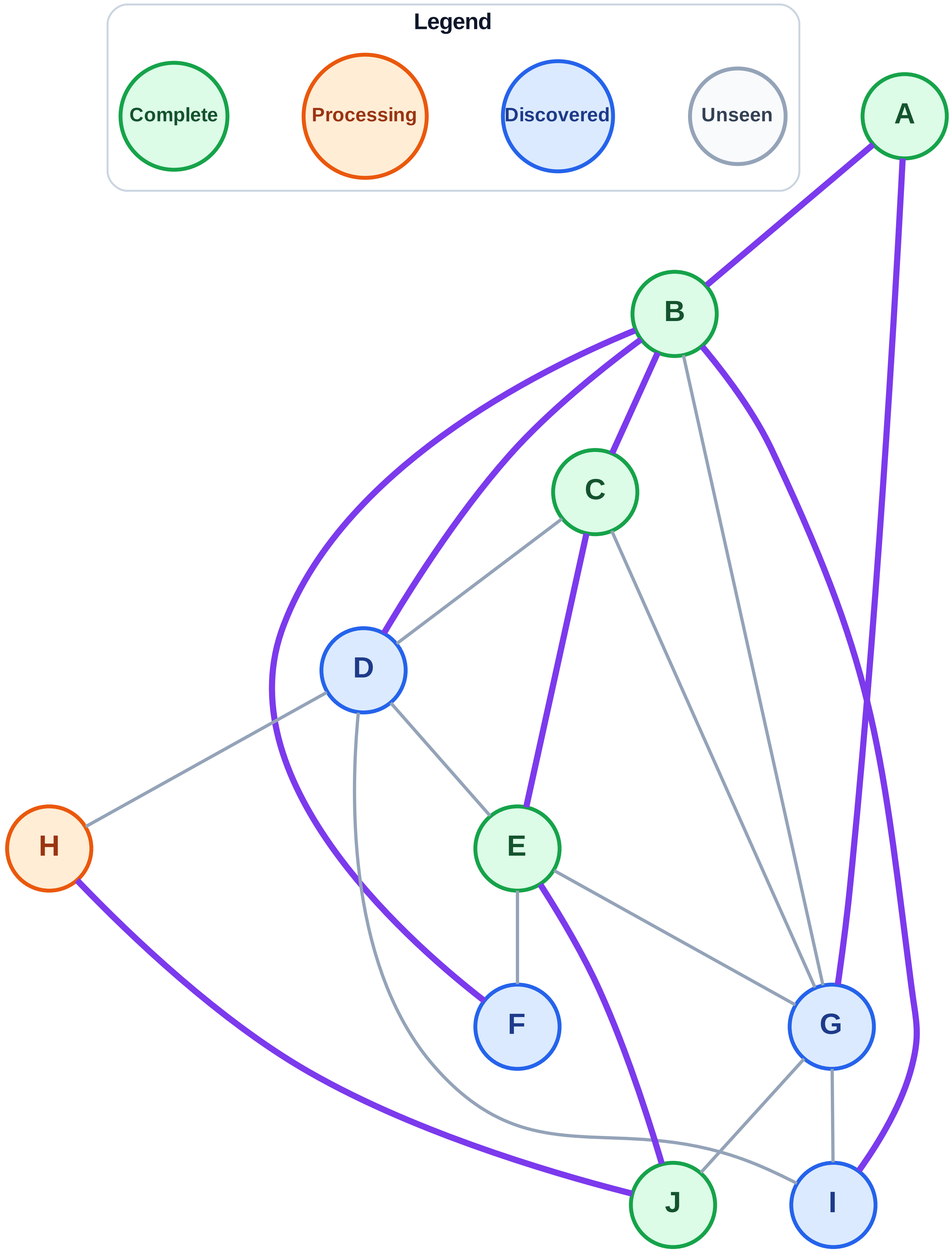
Step 10: Process node J




DFS Traversal
Step 11: Discover H from J
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): G | I | F | D | H
Visited: A, B, C, E, J
A
B
C
D
E
F
G
H
I
J

DFS Traversal

Step 12: Process node H

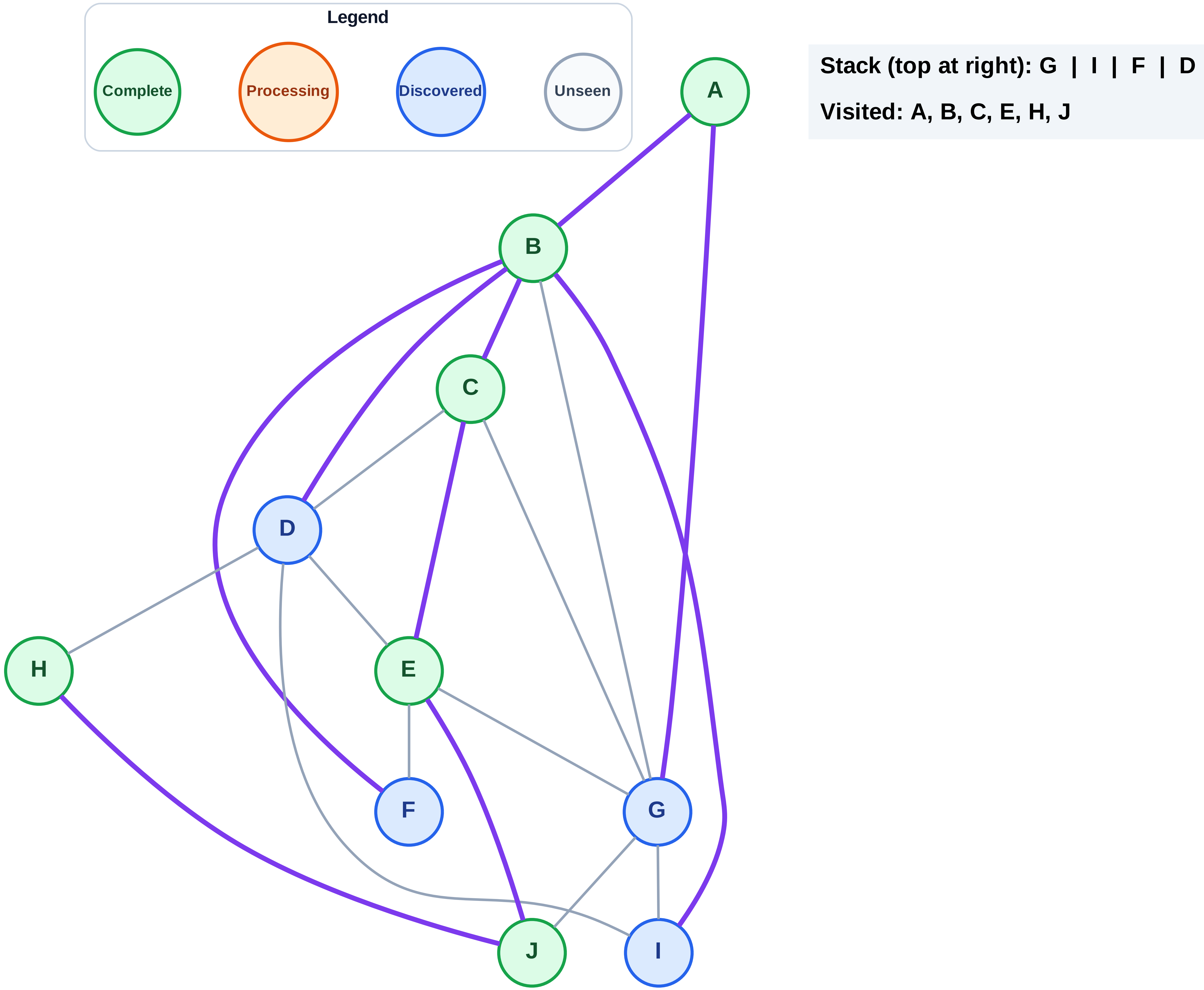


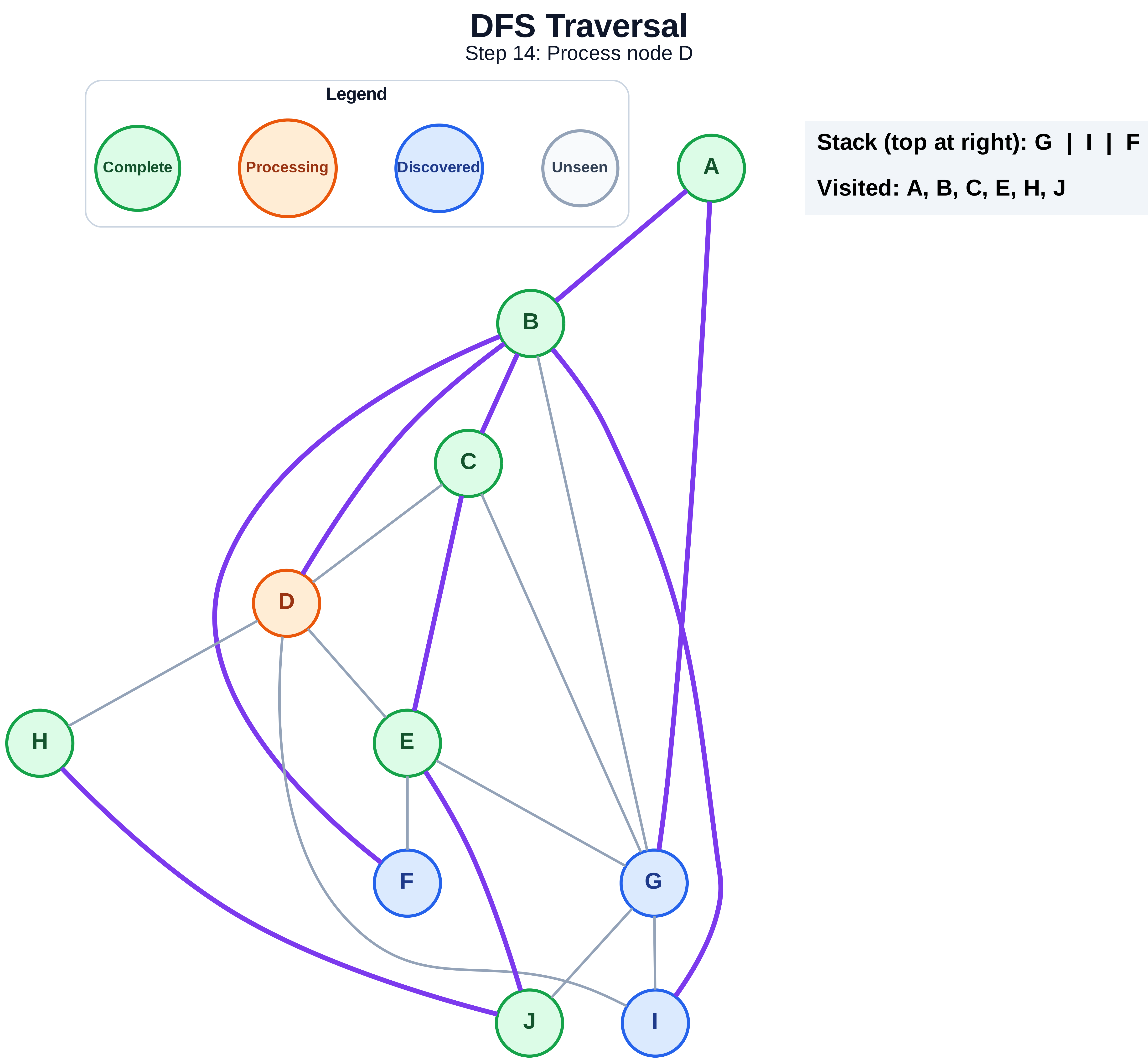
Stack (top at right): G | I | F | D

Visited: A, B, C, E, J

DFS Traversal

Step 13: No new neighbors from H

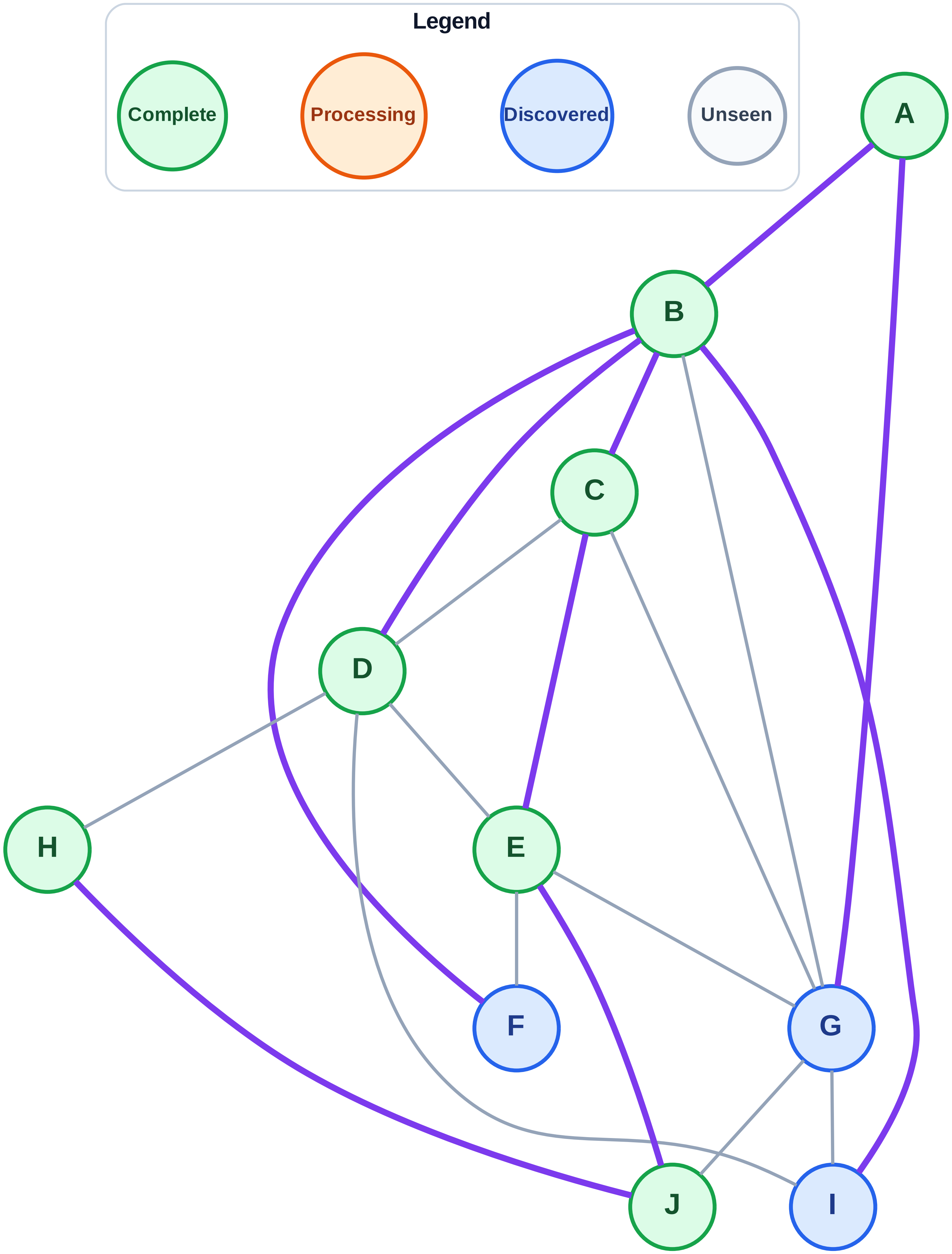




DFS Traversal
Step 14: Process node D
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): G | I | F
Visited: A, B, C, E, H, J
A
B
C
D
E
F
G
H
I
J

DFS Traversal

Step 15: No new neighbors from D

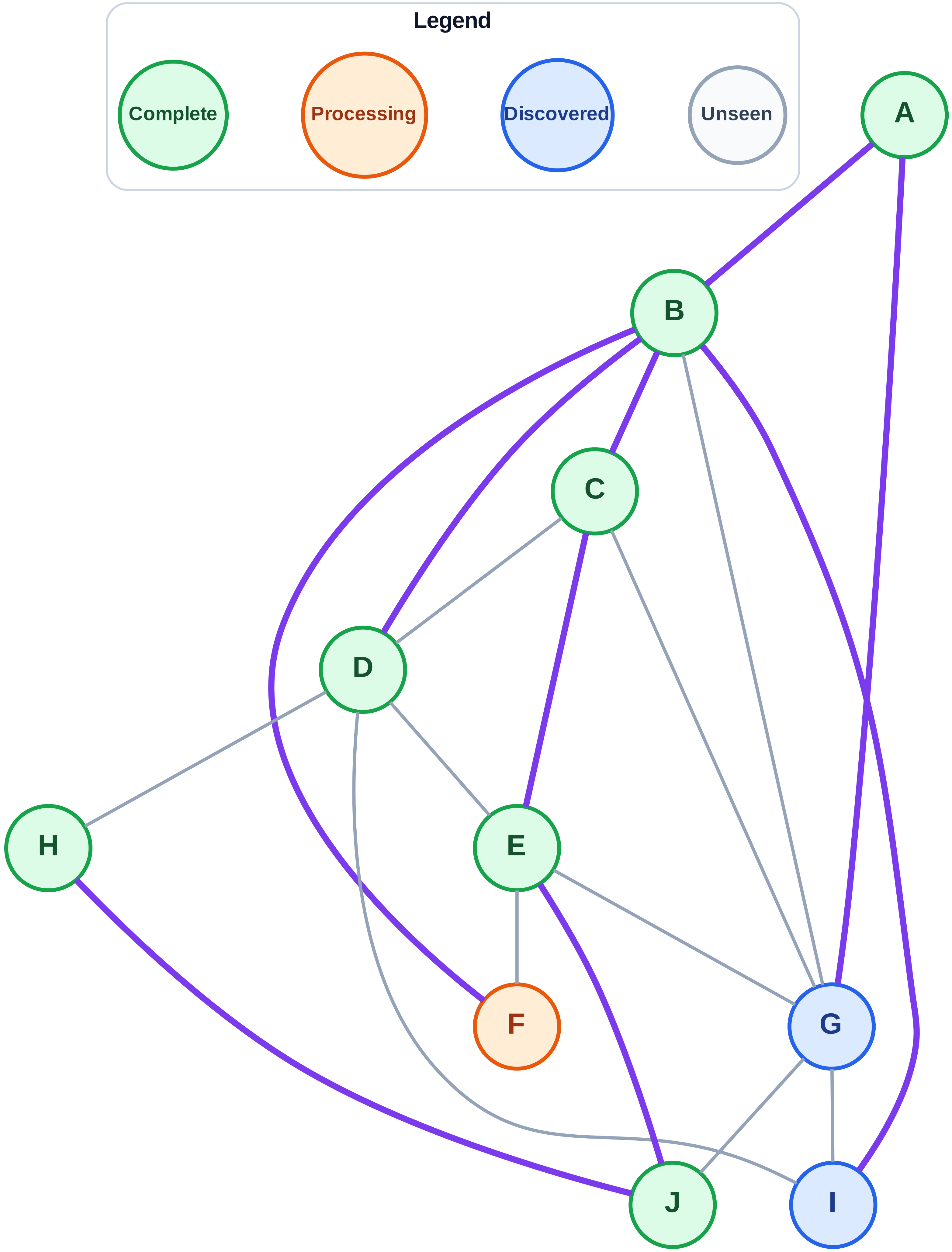


Stack (top at right): G | I | F

Visited: A, B, C, D, E, H, J

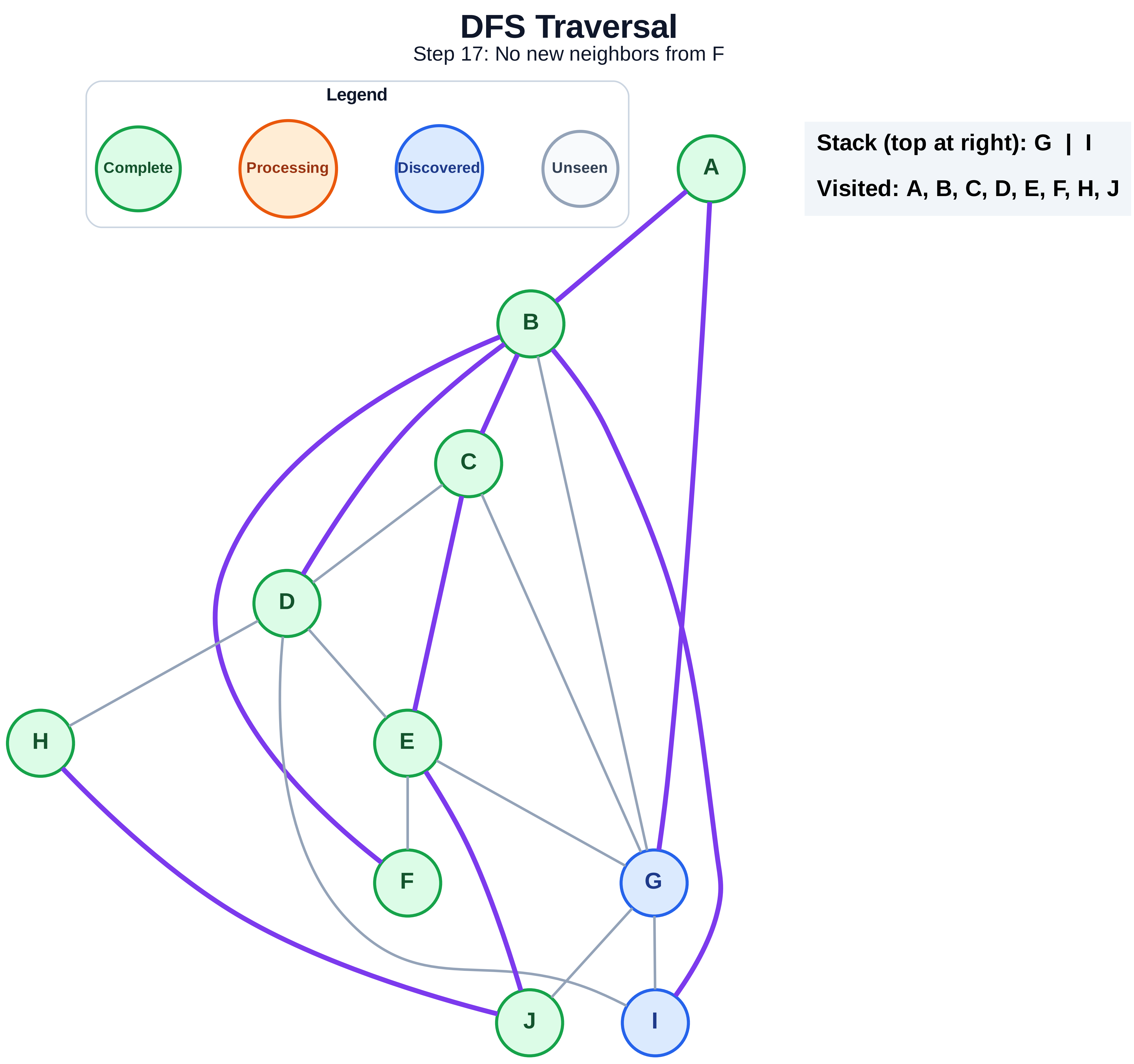
DFS Traversal

Step 16: Process node F



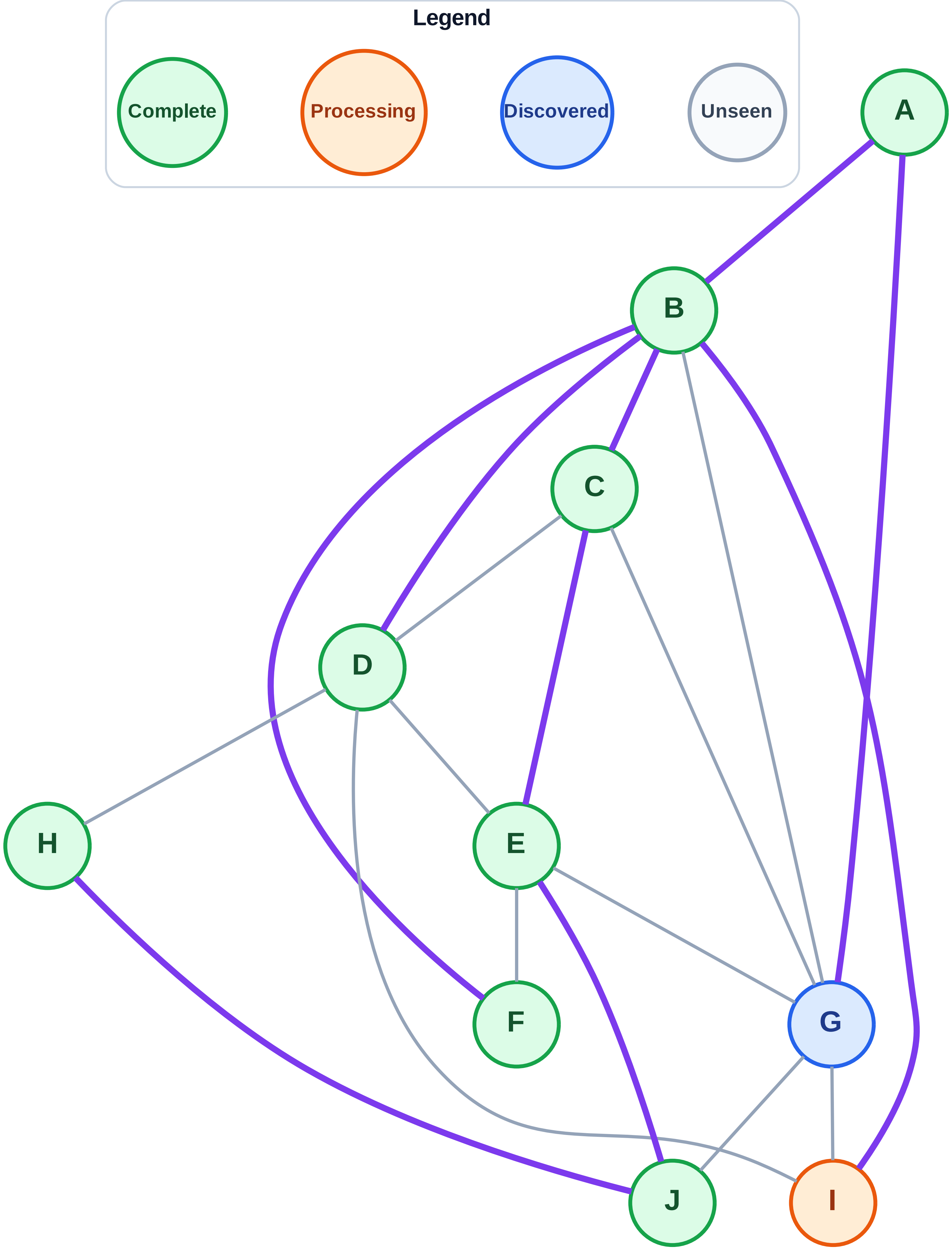
Stack (top at right): G | I

Visited: A, B, C, D, E, H, J


DFS Traversal
Step 17: No new neighbors from F
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): G | I
Visited: A, B, C, D, E, F, H, J
A
B
C
D
E
F
G
H
I
J

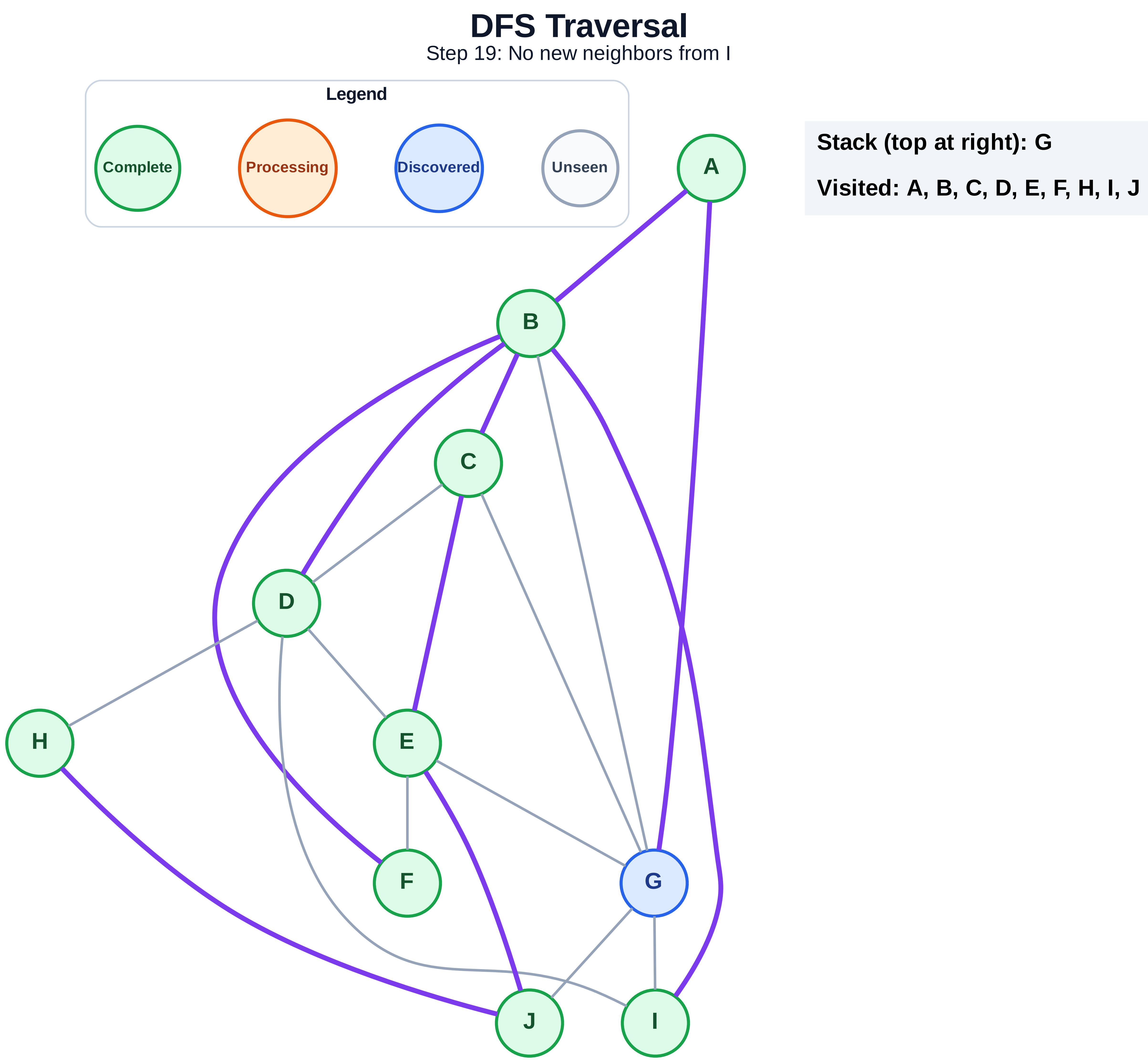
DFS Traversal

Step 18: Process node I



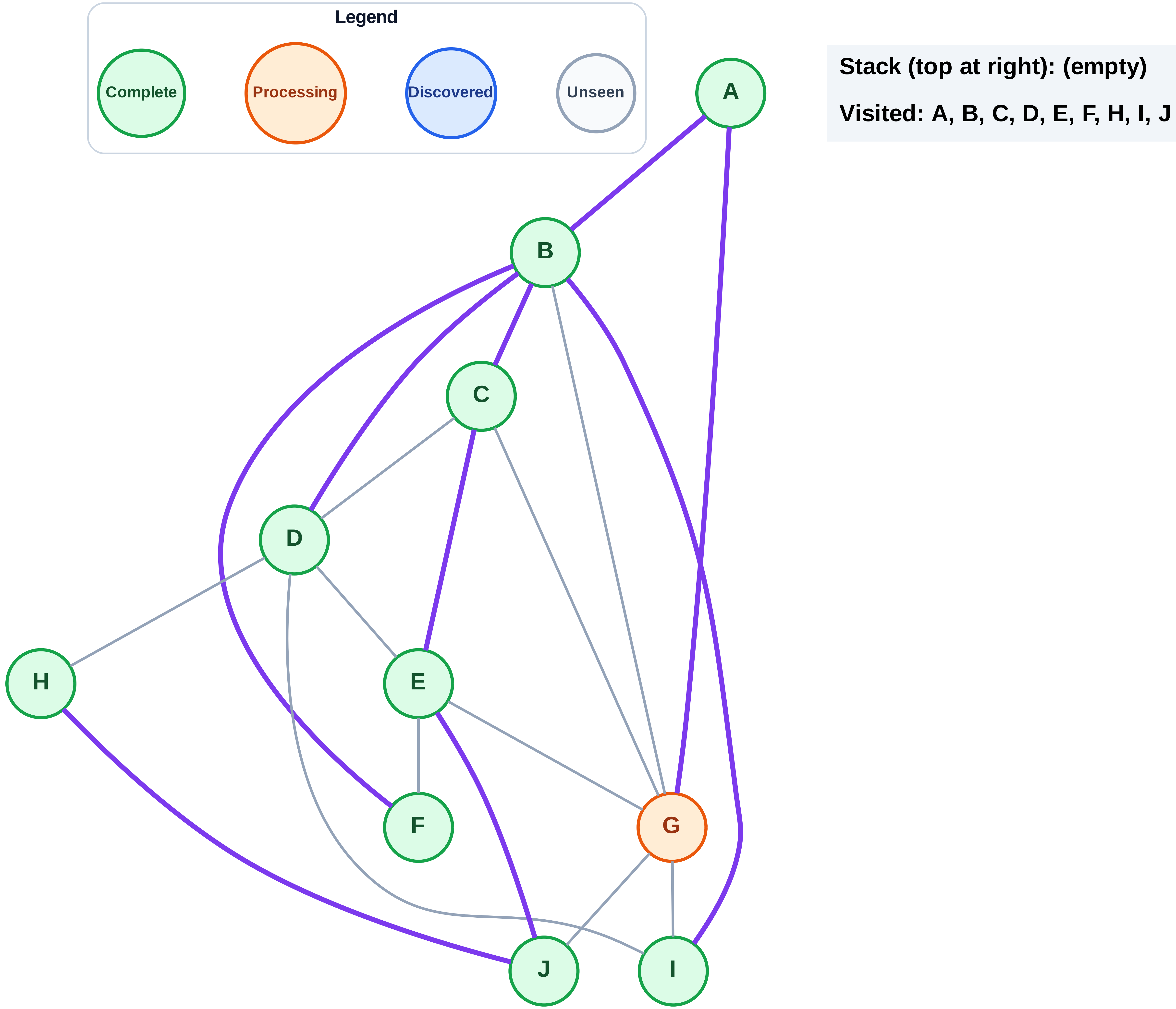
Stack (top at right): G

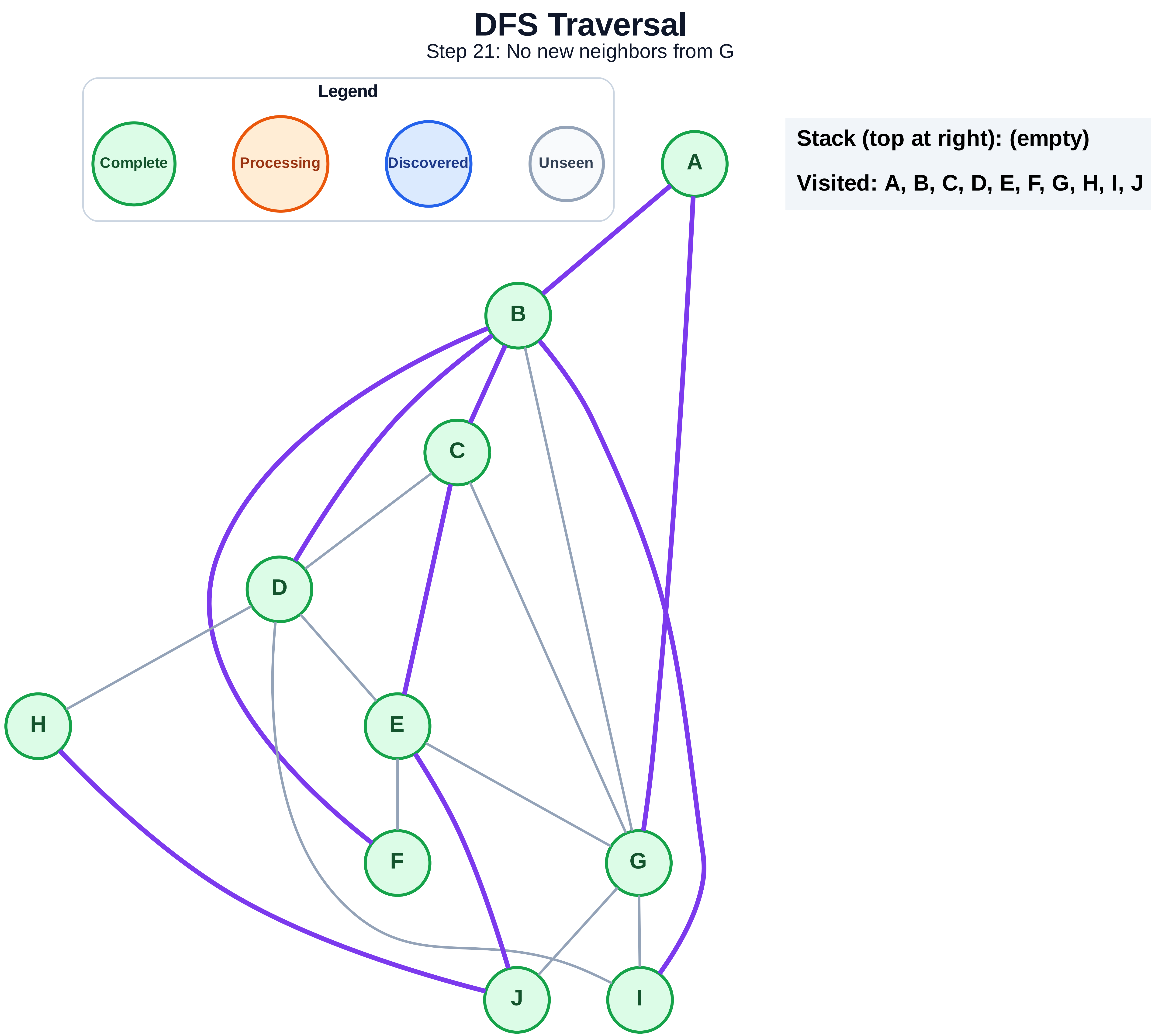
Visited: A, B, C, D, E, F, H, J


DFS Traversal
Step 19: No new neighbors from I
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): G
Visited: A, B, C, D, E, F, H, I, J
A
B
C
D
E
F
G
H
I
J

DFS Traversal

Step 20: Process node G





DFS Traversal
Step 21: No new neighbors from G
Legend
Complete
Processing
Discovered
Unseen
Stack (top at right): (empty)
Visited: A, B, C, D, E, F, G, H, I, J
A
B
C
D
E
F
G
H
I
J